RRB 2025 Exams

SelectionWay

RRB Technician Grade-3

Science and General Awareness

ALL 450 QUESTIONS IN 9 SHIFTS

(English Medium)

Chapter-wise
Hard to Easy Level Order

By
Gagan Pratap



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History (Q-11)

Q1) The Ramosi Revolt was led by which tribal leader in 1822 against Britishers?

A) Chittur Singh **B)** Bakshi Jagabandhu **C)** Veer Surendra Sai **D)** Kittur Chennamma

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q2) Keshab Chandra Sen was a prominent leader of which of the following reform movements in India?

A) Prarthana Samaj **B)** Brahmo Samaj **C)** Arya Samaj **D)** Ramakrishna Mission

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q3) Provincial autonomy in provinces in place of dyarchy was introduced by which of the following British Indian government legislations?

A) Indian Councils Act of 1909 **B)** Government of India Act of 1935 **C)** Indian Councils Act of 1892

D) Government of India Act of 1919

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q4) Who among the following were the pivotal leaders of the Santhal Rebellion of 1855–1856?

A) Tilka Manjhi and Sido Kanhu **B)** Birsa Munda and Raghunath Murmu **C)** Lalu Prasad and Sidhu Kanhu

D) Sidhu Murmu and Kanhu Murmu

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q5) Which of the following British Indian government legislations provided for the establishment of a Supreme Court at Calcutta, comprising one chief justice and three other judges?

A) Amending Act of 1781 **B)** Regulating Act of 1773 **C)** Pitt's India Act of 1784 **D)** Charter Act of 1793

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q6) Which policy of Lord Dalhousie was a significant political cause of the 1857 Revolt?

A) Dual Government **B)** Doctrine of Lapse **C)** Vernacular Press Act **D)** Ryotwari System

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q7) At which session of the Indian National Congress did the party formally split into the Moderates and the Extremists?

A) Delhi session 1912 **B)** Calcutta session 1906 **C)** Bombay session 1910 **D)** Surat session 1907

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q8) Who was chosen as the first president of the Indian National Congress (INC) during its first session?

A) Womesh Chandra Bonnerjee **B)** Surendranath Banerjee **C)** Bipin Chandra Pal **D)** Pherozeshah Mehta

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q9) Which among the following battles in 1757 marked the beginning of British political control in India?

A) Battle of Buxar **B)** Battle of Colachel **C)** Battle of Seringapatam **D)** Battle of Plassey

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q10) Which among the following slogans was popularised by Bal Gangadhar Tilak?

A) Swaraj is my birthright and I shall have it **B)** Give me blood and I will give you freedom

C) Freedom is our goal and we shall achieve it **D)** Liberty is our dream and we must attain it

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q11) The inaugural railway line in the Indian subcontinent was opened in 1853 between between which two locations?

A) Patna and Gaya **B)** Jaipur and Ajmer **C)** Bombay and Thane **D)** Surat and Bharuch

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

SelectionWay

Geography (Q-8)

Q12) In which Indian state do the Western Ghats reach their maximum elevation, marked by the presence of Anaimudi, the highest peak of the range?

A) Maharashtra **B)** Tamil Nadu **C)** Kerala **D)** Karnataka

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q13) With reference to fluvial geomorphic processes in the Indian subcontinent, the Indo-Gangetic Plains have been formed predominantly due to the long-term accumulation of which specific type of deposits?

A) Aeolian sand deposits formed under arid climatic conditions **B)** Glacial till deposited directly by valley glaciers

C) Marine sediments laid down by repeated sea transgressions **D)** Alluvial sediments deposited by Himalayan river systems

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q14) Which of the following is the highest peak of the Western Ghats mountain range?

A) Mullayanagiri, Baba Budan Hills **B)** Anaimudi, Anamalai Hills **C)** Mahabaleshwar, Sahyadri plateau region

D) Doddabetta, Nilgiri Hills

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q15) Which of the following cities experiences the retreat of Monsoon last?

A) Chennai **B)** Mumbai **C)** Jaipur **D)** Ahmedabad

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q16) Which distinctive drainage characteristic best describes the river Luni of north-western India?

A) It flows directly into the Arabian Sea. **B)** It forms a large, well-developed delta at its mouth.

C) It is the only glacier-fed perennial Himalayan river.

D) It terminates in an inland salt marsh in the Rann region without reaching the sea.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q17) The Chota Nagpur Plateau of eastern India is especially known for the abundance of which of the following mineral resources that support the country's industry?

A) Diamond-bearing kimberlite rocks **B)** Petroleum and natural gas reserves **C)** Coal and iron ore deposits

D) Limestone used primarily for cement manufacture

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q18) Cherrapunji and Mawsynram, two places that generally receive the maximum rainfall in India, are located in:

A) Khasi Hills **B)** Nilgiri Hills **C)** Aravalli Hills **D)** Satpura Hills

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q19) India shares a *maritime boundary* (sea boundary) with which one of the following countries across the Arabian Sea or the Indian Ocean?

A) Maldives **B)** Mongolia **C)** Russia **D)** Afghanistan

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Polity (Q-18)

Q20) The Government has proposed to include which Union Territory under the ambit of Article 240 of the Constitution?

A) New Delhi **B)** Chandigarh **C)** Lakshadweep **D)** Puduchery

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q21) Which of the following British Indian government legislation separated for the first time 'provincial budgets' from the 'central budget' and authorised the provincial legislatures to enact their budgets?

A) Government of India Act of 1935 **B)** Indian Councils Act of 1892 **C)** Indian Councils Act of 1909

D) Government of India Act of 1919

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q22) The term 'Armed Rebellion' was introduced under Article 352 of the Indian Constitution for the proclamation of a National Emergency by which Constitutional Amendment?

A) 42nd Constitutional Amendment Act, 1976 **B)** 86th Constitutional Amendment Act, 2002

C) 61st Constitutional Amendment Act, 1988 **D)** 44th Constitutional Amendment Act, 1978

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q23) Which of the following is NOT one of the valid ways to acquire Indian citizenship under the Citizenship Act, 1955?

A) By registration **B)** By marriage **C)** By birth **D)** By descent

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q24) Which of the following Directive Principles was NOT added to the Indian Constitution as part of the 42nd Amendment Act of 1976?

A) Article 48A: To protect and improve the environment and safeguard forests and wildlife

B) Article 39A: To promote equal justice and to provide free legal aid to the poor

C) Article 38: State to minimise inequalities in income, status, facilities and opportunities

D) Article 39: To secure opportunities for healthy development of children

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q25) In September 2025, the Supreme Court of India reaffirmed which important legal position under the Scheduled Castes and the Scheduled Tribes (Prevention of Atrocities) Act, 1989?

A) The grant of anticipatory bail is absolutely barred if the FIR prima facie discloses an offence under the Act

B) The accused can apply for anticipatory bail under all circumstances

C) The anticipatory bail provision has been removed from the Indian Penal Code

D) The anticipatory bail may be granted only by the High Court

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q26) Which Article of the Constitution of India provides for the establishment of the National Commission for Scheduled Castes (NCSC) to safeguard the rights of the Scheduled Castes?

A) Article 330 **B)** Article 350 **C)** Article 338 **D)** Article 340

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q27) Who among the following appoints the members of the State Public Service Commission (SPSC) under Article 316 of the Constitution of India?

A) President of India **B)** Chief Justice of the High Court **C)** Prime Minister of India **D)** Governor of the State

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q28) Allocation of seats to States and Union Territories in the Rajya Sabha is provided under which of the following Schedules of the Constitution of India?

A) Tenth Schedule **B)** Sixth Schedule **C)** Fourth Schedule **D)** Eighth Schedule

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q29) The provision of 'Single citizenship' for all in the Indian Constitution is adopted from the constitution of which of the following countries?

A) US Constitution **B)** British Constitution **C)** Australian Constitution **D)** Canadian Constitution

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q30) Division of legislative subjects into union, state and concurrent List is contained in which of the following schedules of the Constitution of India?

A) Seventh Schedule **B)** Eleventh Schedule **C)** Ninth Schedule **D)** Fifth Schedule

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q31) In India, which of the following is NOT considered as a constitutional body?

A) National Human Rights Commission **B)** Election Commission of India **C)** Union Public Service Commission **D)** Finance Commission of India

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q32) Which of the following articles of the Constitution of India provides for the mechanism and power of Parliament to amend the Constitution?

A) Article 372 **B)** Article 362 **C)** Article 356 **D)** Article 368

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q33) "To provide opportunities for education to his child or ward between the ages of six and fourteen years" was added to the Indian Constitution as the 11th Fundamental Duty through which of the following amendments?

A) 42nd Constitutional Amendment Act, 1976 **B)** 61st Constitutional Amendment Act, 1988 **C)** 44th Constitutional Amendment Act, 1978 **D)** 86th Constitutional Amendment Act, 2002

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q34) Which of the following 1973 landmark judgments of the Supreme Court of India provided that Parliament cannot amend the 'basic structure' of the Constitution through amendments?

A) Champakam Dorairanjan Case **B)** Minerva Mills Case **C)** Kesavananda Bharati Case **D)** Golakhnath Case

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q35) The Constitution of India provided for how many types of emergencies in Part XVIII, from Articles 352 to 360?

A) Three **B)** Four **C)** Six **D)** Five

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q36) Words 'Integrity', 'Socialist' and 'Secular' were added to the Preamble of the Constitution of India by which of the following amendments?

A) 24th Amendment, 1971 **B)** 42nd Amendment, 1976 **C)** 44th Amendment, 1978 **D)** 52nd Amendment, 1985

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q37) Which of the following amendments to the Indian Constitution is regarded as a major milestone in the country's taxation system for introducing the Goods and Services Tax (GST)?

A) 97th Amendment, 2011 **B)** 101st Amendment, 2016 **C)** 100th Amendment, 2015 **D)** 86th Amendment, 2002

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

SelectionWay

Economics (Q-4)

Q38) In which year did India completely eliminate Quantitative Restrictions (QRs) on imports as part of trade liberalization?

A) 1996 **B)** 2005 **C)** 1998 **D)** 2001

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q39) Which of the following committees, in its 2009 report, is known for introducing a widely used method to estimate poverty levels in India?

A) Rangarajan Expert Group **B)** Suresh Tendulkar Committee **C)** A.V. Subramanian Committee **D)** B.N. Yugandhar Committee

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q40) In which year did India sign the World Trade Organization (WTO) Agreement and become a founding member of WTO?

A) 1997 **B)** 1995 **C)** 1991 **D)** 1993

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q41) The Second Five-Year Plan (1956–61) was strategically designed to accelerate India's long-term growth by prioritizing which of the following sectors?

A) Renewable energy expansion **B)** Public health infrastructure **C)** Heavy industries **D)** Village-level artisan development

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

SelectionWay

Arts & Culture (Q-10)

Q42) Which instrument is described as a hand drum used by tribal communities in the hilly areas of Tripura and Meghalaya?

A) Chyabrung **B)** Davandi **C)** Dhak **D)** Dama

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q43) Who among the following is NOT part of the Trinity of Carnatic Music?

A) Tyagaraja **B)** Muthuswami Dikshitar **C)** Shyama Shastri **D)** Swathi Thirunal

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q44) Who among the following made a significant contribution to Bharatanatyam as a performer, teacher, composer and scholar and was awarded Padma Bhushan in 2011?

A) Vempati Chinna Satyam **B)** Guru Bipin Singh **C)** C. V. Chandrashekhar **D)** Kalamandalam Gopi

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q45) Which dance form is performed by men wearing long colorful robes and tall conical caps in Jammu & Kashmir?

A) Chholiya **B)** Hikat **C)** Dumhal **D)** HurkaBaul

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q46) In June 2025, Indian author Banu Mushtaq, in her acclaimed book Heart Lamp, explores which central theme through her vivid portrayal of everyday characters?

A) Political leadership **B)** Caste and religious oppression **C)** Economic reforms **D)** Environmental activism

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q47) Which of the following renowned Indian musicians performed at the Red Fort on the occasion of India's independence in 1947?

A) Pandit Bhimsen Joshi **B)** Pandit Ravi Shankar **C)** Pandit Hariprasad Chaurasia **D)** Ustad Bismillah Khan

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q48) Match the following.

Person	**Musical Instrument**
a. Amir Ali Khan	i. Sitar
b. Anoushaka Shankar	ii. Ghatam
c. T.H. Vinayakram	iii. Sarod
d. Bismillah Khan	iv. Shehnai

A) a-ii; b-iii; c-i; d-iv **B)** a-iii; b-i; c-ii; d-iv **C)** a-ii; b-iv; c-i; d-iii **D)** a-iv; b-ii; c-iii; d-i

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q49) Which of the following dancers is acclaimed for their mastery and teaching of Odissi?

A) Kalamandalam Leelamma **B)** Sitara Devi **C)** Kelucharan Mohapatra **D)** Vempati Chinna Satyam

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q50) Which of the following classical music instruments did Ustad Bismillah Khan mostly popularise?

A) Veena **B)** Shehnai **C)** Sitar **D)** Tabla

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q51) Which Indian classical dance form originated in the southern state of Tamil Nadu and is traditionally performed in temples as a devotional offering?

A) Bharatanatyam **B)** Odissi **C)** Kathak **D)** Mohiniyattam

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

SelectionWay

Current Affairs (Q-25)

Q52) In which state was the 22nd All India TDS Conference organised to review Income Tax Act amendments?

A) Rajasthan **B)** Maharshtra **C)** Gujarat **D)** Madhya Pradesh

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q53) According to the Ministry of Defence, what was India's highest-ever defence production in 2024–25?

A) ₹1.54 Lakh Crore **B)** ₹2.54 Lakh Crore **C)** ₹2.94 Lakh Crore **D)** ₹3.14 Lakh Crore

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q54) Where was the 30th Senior Women's National Football Championship – Rajmata Jijabai Trophy final match played?

A) Bilaspur **B)** Narainpur **C)** Raipur **D)** Raigarh

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q55) During October 2025, what first-ever scientific observation was made by ISRO's Chandrayaan-2 orbiter?

A) The effects of Coronal Mass Ejections from the Sun on the Moon

B) The presence of subsurface water deposits under the lunar surface

C) The discovery of a newly identified mineral present on the Moon

D) The detection of volcanic activity occurring beneath the lunar crust

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q56) Which of the following companies has secured a significant contract to indigenously produce the BvS10 Sindhu, an advanced all-terrain armoured vehicle, for the Indian Army?

A) Mahindra Defence Systems **B)** Bharat Electronics Ltd. (BEL) **C)** Hindustan Aeronautics Ltd. (HAL)

D) Larsen & Toubro (L&T)

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q57) In which country is the military exercise ZAPAD 2025 conducted from 10 to 16 September 2025?

A) Germany **B)** France **C)** Japan **D)** Russia

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q58) PM Modi unveils how many Global Initiatives at the G20 Johannesburg Summit 2025?

A) Five **B)** Seven **C)** Six **D)** Four

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q59) Which country has requested WTO dispute consultations with India regarding certain Indian measures in the automotive and renewable energy sectors?

A) China **B)** Brazil **C)** America **D)** England

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q60) Vice-President CP Radhakrishnan inaugurated the 30th CII Partnership Summit in which city?

A) Visakhapatnam **B)** Patna **C)** New Delhi **D)** Ahmedabad

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q61) Reserve Bank of India (RBI) has raised India's GDP growth forecast for FY26 to .

A) 7.3% **B)** 6.7% **C)** 6.9% **D)** 6.8%

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q62) Which private Indian company has recently achieved a significant milestone by becoming the first to receive approval for offering satellite-based internet services in India?

A) Tata Communications **B)** Ananth Technologies **C)** SpaceX **D)** Reliance Jio

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q63) What is the main purpose of the vehicle-mounted Laser DEW MK-II(A) tested by DRDO in April 2025?

A) Satellite launching **B)** Anti-ballistic missile defense **C)** Space communication **D)** Drone neutralization

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q64) Which Indian won a bronze medal in the women's individual compound event at the Archery World Cup Final 2025 in Nanjing?

A) Ankita Bhakat **B)** Bhajan Kaur **C)** Jyothi Surekha Vennam **D)** Deepika Kumari

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q65) Which bank joins the India Bullion Exchange as a special category client?

A) Bank of India **B)** Indian Bank **C)** Punjab National Bank **D)** State Bank of India

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q66) The recent $131 million deal for cutting-edge surveillance technology from HawkEye 360 involves which two countries ?

A) India and Russia **B)** India and Ukraine **C)** India and United States **D)** India and Japan

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q67) The Supreme Court dismissed which state's plea to halt Karnataka's Mekedatu project in November 2025?

A) Tamil Nadu **B)** Goa **C)** Maharashtra **D)** Andhra Pradesh

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q68) Who became the first Indian senior world champion in speed skating by winning the 1,000 m sprint at the World Championships in September 2025?

A) Krish Sharma **B)** Dhanush Babu **C)** Anandkumar Velkumar **D)** Anish Raj

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q69) Which of the following bodies approved the appointment of S Krishnan as Chairman of J&K Bank?

A) NITI Aayog **B)** Ministry of Finance **C)** Insurance Regulatory and Development Authority of India(IRDAI)

D) Reserve Bank of India

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q70) Which agency has bagged a Transfer of Technology (ToT) deal, valued at ₹511 crore, from the Indian Space Research Organisation, to build and operate Small Satellite Launch Vehicles (SSLVs)?

A) Hindustan Aeronautics Limited **B)** Larsen & Toubro **C)** Adani Defence **D)** Bharat Dynamics Limited

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q71) Indigenous GSAT-7R weighs 4400 kg, making it the heaviest communication satellite launched from which country?

A) United State of America **B)** France **C)** Japan **D)** India

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q72) Which country launched the first indigenous, Low-Cost Gene Therapy for Sickle Cell Disease, named "BIRSA 101"?

A) India **B)** Japan **C)** China **D)** Spain

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q73) The Henley Passport Index 2025 ranks India at which rank globally?

A) 86 **B)** 85 **C)** 84 **D)** 83

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q74) Which of the following successfully conducted user trials of the Akash-NG missile?

A) Mazagon Dock Shipbuilders **B)** Defence Research and Development Organisation (DRDO)

C) Hindustan Aeronautics (HAL) **D)** Bharat Electronics (BEL)

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q75) Who took the oath as Bihar's Chief Minister for the record 10th time in 2025?

A) Shri Nitish Kumar **B)** Tejashwi Prasad Yadav **C)** Lalu Prasad Yadav **D)** Shri Nishant Kumar

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q76) Which major operation did the Indian government launch in May 2025 following the terrorist attack in Pahalgam?

A) Operation Sindoor **B)** Operation Trident **C)** Operation Shakti **D)** Operation Vijay

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Physics (Q-134)

Q77) A wire has a resistance of 10 Ω. If it is stretched to three times its original length (volume remaining constant), its new resistance will be _____.

A) 90 Ω **B)** 3.33 Ω **C)** 10 Ω **D)** 30 Ω

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q78) Select the correct option regarding the given statements.

Statement 1: Suspensions scatter light.
Statement 2: When particles settle down, they no longer scatter light.

A) Statement 1 is true and Statement 2 is false. **B)** Both statements are false.

C) Both statements are true, and Statement 2 explains Statement 1.

D) Both statements are true, but Statement 2 does not explain Statement 1.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q79) A charged particle moves perpendicular to the axis of a circular current loop. The force on the particle is _____.

A) Maximum at center **B)** Depends on position and velocity **C)** Constant everywhere **D)** Always zero

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q80) The extent of bending of light rays achieved by a lens is expressed in terms of its _____.

A) Image Distance **B)** Aperture **C)** Magnification **D)** Power

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q81) Read the given assertion and reason and select the correct option based on the analysis of these statements.

Assertion: Stethoscope uses the concept of multiple reflection of sound waves for its working.
Reason: Sound waves require a material medium for their propagation.

A) Both assertion and reason are true and the reason is the correct explanation of the assertion.

B) Assertion is false and reason is true.

C) Both assertion and reason are true and the reason is not the correct explanation of the assertion.

D) Assertion is true but reason is false.

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q82) Which of the following is/are NOT true about the distance-time graph for a body moving in uniform motion?

(i) The slope of the distance-time graph gives the acceleration of the body.
(ii) The lesser the slope of the graph the slower the body is moving.
(iii) The area under this graph gives the speed of the body.

A) Only (iii) **B)** Both (i) and (iii) **C)** Only (ii) **D)** Both (i) and (ii)

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q83) A 2.0 cm tall object is placed 20 cm from a convex lens of focal length 15 cm. What is the magnification produced by the lens?

A) +1 **B)** -3 **C)** - 1 **D)** +3

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q84) Reshma swims back and forth in a 120 m long pool, covering a total distance of 240 m in 1 minute. What is her average velocity?

A) 0 m/s **B)** 4 m/s **C)** 5 m/s **D)** 2 m/s

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q85) Which of the following statements is/are correct regarding Rutherford's model of the atom?

Statement 1: The atom has a positively charged dense nucleus at its centre.

Statement 2: Electrons revolve in circular orbits around the nucleus without losing energy.

Statement 3: Most of the space inside the atom is empty.

A) Only 1, 2 and 3 **B)** Only 1 and 3 **C)** Only 2 and 3 **D)** Only 1 and 2

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q86) Which of the following statements is/are INCORRECT about the loudness and softness of a sound wave?
(i) A loud sound has more amplitude as compared to a soft sound.
(ii) A loud sound has lesser energy associated with it as compared to a soft sound.
(iii) Loudness of sound wave is the measure of the response of our heart to the sound.

A) Both (ii) and (iii) **B)** Only (i) **C)** Both (i) and (iii) **D)** Only (ii)

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q87) An object is placed at 100 cm on the left of a convex lens of focal length 40 cm. Where should a screen be placed to right of the lens so that the image formed is real, inverted and smaller in size as compared to the object size?

A) The screen should be placed at 40 cm from the lens.

B) The screen should be placed anywhere between 20 cm and 40 cm from the lens.

C) The screen should be placed anywhere between 40 cm and 80 cm from the lens.

D) The screen should be beyond 100 cm from the lens.

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q88) Which statement best describes the arrangement of electrons in Thomson's atomic model?

A) Electrons revolve around the nucleus **B)** Electrons are found only on the surface of the atom

C) Electrons are embedded in a positively charged sphere **D)** Electrons are located at the center of the atom

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q89) A convex mirror used as a rear-view mirror in a car has a radius of curvature of 4.00 m. A bus is located 6.00 m in front of the mirror. What will be the position and nature of the image formed?

A) Image is formed at 1.5 m behind the mirror **B)** Image is formed at 1.2 m behind the mirror

C) The image is formed at 3.0 m behind the mirror **D)** Image is formed at 3.0 m in front of the mirror

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q90) A train starting from rest reaches a speed of 40 km/h in 10 minutes. What is its final speed in m/s?

A) 4 m/s **B)** 400 m/s **C)** 6.67 m/s **D)** 11.11 m/s

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q91) Newton's first law of motion explains the concept of _____.

A) Continuous acceleration **B)** Balanced forces **C)** Change in momentum **D)** Net force being non-zero

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q92) A block of mass 20 kg is dropped from the top of a building of height 200 m. What will be the velocity with which the block will hit the ground? (Take the value of g = 10 m/s^2)

A) $20\sqrt{5}$ m/s **B)** 40 m/s **C)** $20\sqrt{10}$ m/s **D)** 20 m/s

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q93) A car has a mass of 120 kg. How much work must be done to raise its speed from 72 km/h to 108 km/h?

A) 3.0×10^4 J **B)** 4.0×10^4 J **C)** 1.0×10^4 J **D)** 2.0×10^4 J

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q94) An echo is heard in 6 s. If the distance of the reflecting surface from the source is 990 m, then how much is the speed of sound?

A) 660 m/s **B)** 165 m/s **C)** 330 m/s **D)** 1320 m/s

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q95) The far point of a myopic person is 80 cm in front of the eye. Assuming the far point for normal vision is infinity, what is the required power (P) of the corrective lens?

A) +0.80 D **B)** +1.25 D **C)** -0.80 D **D)** -1.25 D

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q96) Which of the following is/are NOT true about action reaction forces?

(i) Action reaction forces are unequal in magnitude and always act in the same direction.
(ii) Action reaction forces act on different bodies.
(iii) Action reaction forces act on the same body.

A) Both (i) and (iii) **B)** Only (ii) **C)** Only (iii) **D)** Both (i) and (ii)

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q97) A body moves 6 km towards the east in 1 hour and then 8 km towards the west in the next 1 hour. The average velocity of the body for the whole journey is _____ .

A) 0 **B)** 7 km/h east **C)** 1 km/h east **D)** 1 km/h west

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q98) A convex mirror of focal length 'X' cm forms an image at a distance of 8 cm when the object is placed at a distance of 40 cm in front of the mirror. What is the value of X?

A) - 10 cm **B)** + 10 cm **C)** - 20/3 cm **D)** + 20/3 cm

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q99) The speed of sound depends primarily on the nature and _____ of the medium.

A) Volume **B)** Pressure **C)** Temperature **D)** Amplitude

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q100) A bus moves from one end A to the other end B of a straight 1000 m road in 100 s and then turns around and moves 500 m back to point C in another 100 s. What will be the average velocity (in m/s) and the type of motion of the bus in going from point A to point C?

A) 0.4 m/s and Non-Uniform Motion **B)** 0.4 m/s and Uniform Motion **C)** 2.5 m/s and Non-Uniform Motion

D) 2.5 m/s and Uniform Motion

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q101) When white light is incident obliquely on a glass prism, which of the following statement(s) is/are NOT true?

(i) Red colour undergoes minimum deviation.
(ii) The speed of violet colour is the least while travelling through the prism.
(iii) Refractive index of green colour is more than the violet colour.
(iv) Refractive index of red colour is more than blue colour.

A) Both (i) and (ii) **B)** Both (i) and (iv) **C)** Both (ii) and (iii) **D)** Both (iii) and (iv)

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q102) A concave lens has a focal length of 15 cm. If it forms an image 10 cm from the lens, how far away is the object placed from the lens?

A) -25 cm **B)** -30 cm **C)** +30 cm **D)** +6 cm

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q103) A convex mirror produces a magnification of 1/3 when an object is placed at a distance of 30 cm from it. How much will be the image distance?

A) - 90 cm **B)** + 10 cm **C)** + 90 cm **D)** - 10 cm

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q104) At high altitudes such as hill stations, water boils at a temperature lower than 100°C, whereas in a pressure cooker it boils above 100°C. Which of the following statements correctly explains this phenomenon?

A) Boiling point of a liquid increases with an increase in atmospheric pressure and decreases with a decrease in atmospheric pressure.

B) Boiling point of a liquid decreases with an increase in atmospheric pressure and increases with a decrease in atmospheric pressure.

C) Boiling point of a liquid is independent of external pressure and depends only on temperature.

D) Water molecules at high altitude have lower kinetic energy than at sea level, so they boil at a lower temperature.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q105) If a velocity–time graph shows a horizontal straight line, the object is moving with _____.

A) No velocity **B)** Constant velocity **C)** Non-uniform acceleration **D)** Zero displacement

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q106) A sound wave has a frequency of 2 kHz and a wavelength of 45 cm. How much time will it take to cover a distance of 1.8 km?

A) 2 seconds **B)** 0.5 seconds **C)** 1 second **D)** 4 seconds

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q107) The distance between the pole and the position of the object placed in front of a concave mirror is 120 cm. A real and inverted image is formed by this mirror on a screen at 40 cm from the pole of this mirror. What will be the focal length of this mirror?

A) - 60 cm **B)** + 30 cm **C)** - 30 cm **D)** + 60 cm

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q108) Why does the sky appear dark to passengers flying at very high altitudes?

A) Scattering of light is not prominent at such heights. **B)** They are above the atmosphere.

C) The path of light is scattered back towards the sun. **D)** The air molecules absorb all blue light at that altitude.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q109) Two bodies A and B are travelling in uniform motion, the slope of the distance- time graph of body A is greater than the slope of the distance - time graph of body B. Which of the following statements is correct?

A) The speed of body A is equal to the speed of body B. **B)** The speed of body A is greater than speed of body B.

C) The speed of body A is less than the speed of body B.

D) The acceleration of body A is greater than the acceleration of body B.

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q110) An ultrasonic wave has a frequency of 33 kHz. What will be the speed of this wave if its wavelength is 0.01 m?

A) 3.3 m/s **B)** 33 m/s **C)** 330 m/s **D)** 0.33 m/s

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q111) If a lens forms a real and inverted image, the height of the image h' is generally taken as _____.

A) Positive, but less than the object height **B)** Positive, as it is above the axis **C)** Negative, only if the object is virtual

D) Negative, as it is below the axis

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q112) Which of the following statements about Thomson's model of an atom is/are correct?

Statement 1. An atom consists of a positively charged sphere with electrons embedded in it.

Statement 2. The mass of the atom is due to neutrons located in the nucleus.

Statement 3. The model is often compared to a watermelon with seeds representing electrons.

A) Only 1, 2, and 3 **B)** Only 2 and 3 **C)** Only 1 **D)** Only 1 and 3

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q113) Two lens of power (+ 4 D) and (- 6 D) are placed in contact with each other. What will be their combined focal length in cm?

A) + 0.5 cm **B)** - 50 cm **C)** - 10 cm **D)** + 0.1 cm

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q114) Vibration, which produces sound, is defined as a kind of rapid _____.

A) Single directional displacement **B)** Linear forward motion **C)** To and fro motion of an object **D)** Slow lateral motion

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q115) For an object moving with uniform velocity, the height of its velocity-time graph remains constant with time. This results in a graph that is a _____.

A) Straight line passing through the origin **B)** Straight line parallel to the x-axis **C)** Vertical line **D)** Curve

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q116) If the speed of light in a medium is 2.25×10^8 m/s, its refractive index is _____.

A) 0.75 **B)** 1.33 **C)** 1.75 **D)** 1.50

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q117) If an object is placed between F_1 and $2F_1$ of a convex lens, the image produced is _____. (F_1 is the position of the principal focus.)

A) Enlarged, real, inverted **B)** Same size, real, inverted **C)** Diminished, real, inverted **D)** Enlarged, virtual, erect

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q118) An object is launched straight upward and reaches a maximum height of 1000 cm. What was the initial velocity of the object? (Take g = 9.8 m/s^2)

A) 6 m/s **B)** 2 m/s **C)** 14 m/s **D)** 10 m/s

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q119) What happens to a ray of light that either passes through the principal focus (for a convex lens) or appears to meet at the principal focus (for a concave lens) before striking the lens?

A) It always converges at the optical centre. **B)** It forms an image at the focal plane of the lens.

C) It emerges parallel to the principal axis after refraction. **D)** It bends randomly after refraction.

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q120) Which quantity decreases when a solid object is immersed in water?

A) Mass **B)** Buoyant force **C)** Apparent weight **D)** Density

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q121) Multiple reflection of sound can be used to _____.

A) Hear sounds clearly over long distances **B)** Reduce loudness **C)** Increase the frequency of sound **D)** Change the pitch of sound

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q122) Six resistors of 3 Ω each are connected in parallel in a circuit. What will be their effective resistance?

A) 18 Ω **B)** 3 Ω **C)** 2 Ω **D)** 0.5 Ω

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q123) A strong magnetic field produced inside a solenoid can be used to magnetise a piece of soft iron placed inside the coil. The resulting magnet is called a/an _____.

A) Toroid **B)** Permanent magnet **C)** Natural magnet **D)** Electromagnet

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q124) If a sample of air is compressed to half its volume, which of the following statements is correct?

A) The composition of air remains same because it is a mixture. **B)** Air becomes a compound on compression. **C)** The composition of air changes. **D)** The proportion of gases changes.

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q125) A distance–time graph for uniform motion must pass through _____.

A) (0, 1) **B)** (1, 1) only **C)** Any point **D)** The origin

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q126) A device has a resistance of 75 Ω and carries a current of 4 A. The power dissipated is _____.

A) 1400 W **B)** 1200 W **C)** 300 W **D)** 18.75 W

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q127) A train starts from rest and reaches a speed of 216 km/h in 10 minutes. If its acceleration is uniform, what is the value of the acceleration?

A) 1.0 m/s2 **B)** 0.6 m/s2 **C)** 0.1 m/s2 **D)** 0.4 m/s2

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q128) A concave lens forms a virtual image when the object is placed _____.

A) At infinity only **B)** At 2F only **C)** At any position in front of the lens. **D)** Beyond 2F only

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q129) Which of the following statements is/are true regarding the speed of sound?
(i) The speed of sound decreases with an increase in the temperature of the medium.
(ii) The speed of sound in solid state is greater than the speed of sound in gaseous state under the same external conditions.
(iii) The speed of sound is independent of the properties of the medium through which it travels.

A) Both (i) and (ii) **B)** Both (i) and (iii) **C)** Only (iii) **D)** Only (ii)

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q130) An object moving at 25 m/s is brought to rest in 5 seconds. The final velocity is _____.

A) 20 m/s **B)** 0 m/s **C)** 25 m/s **D)** 5 m/s

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q131) If the force of gravitation between the Earth and the Moon is 'F' N, then how much will be the force between the Earth and the Moon if the distance between them is doubled? (Keep all the other parameters the same.)

A) 4F **B)** 2F **C)** F/2 **D)** F/4

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q132) When several rays parallel to the principal axis fall on a concave mirror, how do the reflected rays behave?

A) They spread out and appear to come from behind the mirror. **B)** They are reflected back along the same path.

C) They travel parallel to the principal axis after reflection. **D)** They all meet at a single point on the principal axis.

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q133) An unbalanced force of 100 N acts on a body at rest such that it acquires a velocity of 5 m/s in 10 s. How much is the mass of the body?

A) 2 kg **B)** 200 kg **C)** 5 kg **D)** 50 kg

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q134) A body of mass 10 kg is thrown vertically upwards with a velocity of 10 m/s. How much potential energy will be possessed by the body when it reaches the maximum height? (Take g = 10 m/s^2)

A) 500 J **B)** 25 J **C)** 2.5 J **D)** 250 J

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q135) For a body moving with uniform acceleration, average velocity equals _____.

A) Zero **B)** (Initial velocity + Final velocity)/2 **C)** Initial velocity **D)** Final velocity

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q136) If the radius of the Earth were to double while mass remains the same, the new value of g would be _____the original value.

A) 4 times **B)** One-fourth **C)** 2 times **D)** Half

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q137) Which of the following is true about Fleming's left-hand rule?

A) If the thumb points in the direction of current and the middle finger in the direction of magnetic field, the forefinger shows the direction of force.

B) It is used only in electric circuits and not in devices like motors or loudspeakers.

C) If the thumb points in the direction of force and the forefinger in the direction of current, the middle finger shows the direction of magnetic field.

D) If the forefinger points in the direction of magnetic field and the middle finger in the direction of current, the thumb shows the direction of motion or force on the conductor.

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q138) Which of the following statement(s) is/are true regarding the magnetic field line pattern around a straight current carrying wire?

(i) The magnetic field lines around a current carrying wire are in the form of concentric circles.
(ii) The magnetic field lines around a current carrying wire intersect at various points.
(iii) The magnetic field lines move farther apart as one moves away from the wire.

A) Only (i) **B)** Both (ii) and (iii) **C)** Both (i) and (iii) **D)** Only (ii)

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q139) If the gravitational force experienced by a body of mass 'm' on the surface of Earth is 'X' N. If the mass of the body is tripled, what will be the value of the gravitational force experienced by the body?

A) X/9 N **B)** X/3 N **C)** X N **D)** 3X N

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q140) When sound propagates through air, the region of high pressure created by the vibrating object pushing air forward is called a _____.

A) Medium boundary **B)** Transverse wave **C)** Rarefaction (R) **D)** Compression (C)

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q141) The alpha particles that passed straight through the gold foil showed that:

A) the nucleus is negatively charged **B)** the atom is mostly empty space **C)** the nucleus is huge

D) the atom has a positive charge

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q142) In a prism, the emergent ray comes out at an angle different from the incident ray. What is this angle called?

A) Angle of deviation **B)** Angle of incidence **C)** Angle of reflection **D)** Angle of the prism

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q143) A wire carrying current is placed between the poles of a magnet. It experiences maximum force when it _____.

A) is perpendicular to the magnetic field **B)** is outside the magnetic field **C)** is parallel to the magnetic field

D) is at 45° to the magnetic field

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q144) What is the primary reason curved ceilings are used in concert halls?

A) To decorate the hall **B)** To block noise **C)** To absorb sound **D)** To reflect sound evenly

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q145) Even though action and reaction forces are equal in size, they may not cause equal accelerations. Why?

A) Because the forces act on the same object **B)** Because reaction forces are always weaker in effect

C) Because each force acts on a different object, and the objects may have different masses

D) Because only the larger force causes acceleration

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q146) A device often used in electric circuits to change the resistance without changing the voltage source is the _____.

A) Fuse **B)** Ammeter **C)** Voltmeter **D)** Rheostat

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q147) Thomson's model is often referred to as the 'plum pudding model'. In this analogy, what does the 'plum' represent?

A) Neutrons **B)** Protons **C)** Electrons **D)** Alpha particles

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q148) The effect of thrust on a surface depends on the area on which it acts. Force acting on a smaller area exerts a _____ compared to a larger area.

A) Zero pressure **B)** Smaller pressure **C)** Smaller thrust **D)** Larger pressure

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q149) The motion of an athlete running along a circular path with constant speed is an example of accelerated motion because _____.

A) The mass of the athlete is constant **B)** The athletes speed is increasing **C)** A constant external force is applied

D) The direction of motion changes continuously

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q150) A farmer walks along the boundary of a square field with side 11 m, completing one round in 20 seconds. After walking for 3 minutes and 20 seconds, what is the magnitude of the displacement from his starting point?

A) 11 m **B)** 15.56 m **C)** 22 m **D)** 0 m

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q151) The SI unit of force (F) is _____ .

A) N m **B)** Joule **C)** kg m/s^2 **D)** kg m/s

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q152) The location of an object is always described by specifying a point of reference known as the _____.

A) Origin **B)** Displacement **C)** Final position **D)** Axis

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q153) A 90 W electric bulb does how much work in 10 seconds?

A) 900 J **B)** 9 J **C)** 90 J **D)** 9000 J

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q154) In Fleming's Left Hand Rule, which finger represents the direction of the magnetic field?

A) Little finger **B)** Thumb **C)** First (index) finger **D)** Middle finger

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q155) If the velocity-time graph is curved (non-linear), the object is moving with _____.

A) Constant speed **B)** Uniform acceleration **C)** Constant rate of change of momentum **D)** Non-uniform acceleration

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q156) A car of mass 1000 kg moving with a velocity of 20 m/s comes to rest in 10 s after applying the brakes. What is the value of the force applied by the brakes?

A) - 2000 N **B)** - 5000 N **C)** - 500 N **D)** - 200 N

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q157) A wooden block of mass 50 kg is placed on a smooth horizontal surface. A constant horizontal force of 'X' N is applied on it while it is at rest. The block acquires a velocity of 10 m/s in 2 s. What is the value of X?

A) 100 N **B)** 10 N **C)** 25 N **D)** 250 N

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q158) An object is placed at infinity in front of a concave lens. The image formed will be _____ .

A) Real, erect, and at the principal focus **B)** Virtual, erect, and highly diminished at the focus **C)** Virtual, erect, and enlarged

D) Real, inverted, and at infinity

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q159) When applying the equations of motion to an object moving under gravitational force and thrown vertically upwards, the acceleration (a) should be taken as _____.

A) Positive, as gravity is constant. **B)** Positive, as it is in the direction of velocity.

C) Negative, as it opposes the initial motion. **D)** Independent of the direction of motion.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q160) A body travelling in a straight line with a constant velocity of 50 m/s. What is the displacement of the car in 10 sec and the type of motion followed by the car, respectively?

A) 5 m and Uniform Motion **B)** 5 m and Non-Uniform Motion **C)** 500 m and Uniform Motion

D) 500 m and Non-Uniform Motion

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q161) If mass is doubled and g remains constant, weight becomes _____.

A) Half **B)** Four times **C)** Double **D)** Same

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q162) The kinetic energy of an object of mass m moving with a velocity of 5 m/s is 25 J. If its velocity is doubled to 10 m/s, what will be its kinetic energy?

A) 50 J **B)** 100 J **C)** 125 J **D)** 200 J

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q163) The principal focus of a concave mirror is:

A) The point where rays parallel to the principal axis converge after reflection **B)** The pole of the mirror

C) The center of curvature of the mirror **D)** A point on the mirror where all rays meet

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q164) When standing still with a heavy load on your head, you expend energy and get tired. Why is zero work done on the load?

A) The force applied is vertical. **B)** The time elapsed is irrelevant. **C)** There is no displacement of the load.

D) Force is not being applied.

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q165) Which of the following is NOT an application of the Archimede's Principle?

A) Speedometers **B)** Hydrometers **C)** Lactometers **D)** Designing of submarines

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q166) A concave mirror produces a virtual, erect and magnified image when the object is positioned _____.

A) Between the centre of curvature and the focus **B)** Beyond the centre of curvature **C)** Between the focus and the pole

D) At the principal focus

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q167) Which of the following is NOT explained by the Universal Law of Gravitation?

A) Formation of tides **B)** Weight of objects on Earth **C)** Resistance offered by air to a moving object

D) Falling of apples from a tree

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q168) The magnitude of force on a conductor in a magnetic field increases when _____.

A) Magnetic field decreases **B)** Current increases **C)** Length decreases **D)** Current decreases

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q169) What will be the wavelength of a sound wave whose frequency is 220 Hz and speed is 440 m/s in a given medium?

A) 440 m **B)** 220 m **C)** 2 m **D)** 0.5 m

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q170) Two resistors of 5 Ω and 15 Ω are connected in series to a battery of 10 V. How much will flow through the circuit?

A) 0.5 A **B)** 200 A **C)** 2 A **D)** 0.4 A

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q171) The SI unit of the universal gravitation constant is _____.

A) Nm2kg-2 **B)** Nm-2kg-1 **C)** Nm2kg-1 **D)** Nm-2kg2

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q172) If 12 J of work is done to move a charge of 2 C between two points, what is the potential difference between these points?

A) 6 V **B)** 24 V **C)** 14 V **D)** 10 V

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q173) An object travels 16 m in 4 s and then another 16 m in 2 s. What is the average speed of the object?

A) 5.33 m/s **B)** 2.67 m/s **C)** 8 m/s **D)** 4 m/s

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q174) Which of the following statement(s) is/are true when a beam of white light is incident on a glass prism?

(i) The white light undergoes dispersion.
(ii) The speed of light remains constant when it goes from air into the prism.
(iii) The speed of light decreases as the light goes from air into the prism.

A) Only (ii) **B)** Both (i) and (iii) **C)** Only (i) **D)** Both (ii) and (iii)

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q175) How much work is required to move a charge of 3 C between two points that have a potential difference of 15 V?

A) 21 J **B)** 45 J **C)** 14 J **D)** 5 J

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q176) Unbalanced forces acting on an object always cause _____.

A) Increase in mass **B)** Object to remain at rest **C)** No change in motion **D)** Change in motion

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q177) A girl weighing 420 N climbs a rope to a height of 6 m and takes 20 seconds to do it. What is her power output?

A) 126 W **B)** 105 W **C)** 155 W **D)** 90 W

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q178) A body covers a straight-line distance of 10 m from point A to B in 5 s and then covers back the distance from point B to point A along the same straight path in 5 s. How much will be the average speed of the body?

A) 2 m/s **B)** 0.5 m/s **C)** 5 m/s **D)** 200 m/s

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q179) Which of the following situations shows zero acceleration?

A) A train speeding up **B)** An aeroplane taking off **C)** A car moving with constant speed on a straight road **D)** A ball thrown upward

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q180) The centre of curvature (C) of a concave mirror is located _____.

A) Behind the mirror **B)** At the pole (P) **C)** In front of the mirror **D)** At infinity

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q181) Which of the following relations is **incorrect** for electric power (P)?

A) P=I2R **B)** P=R/V2 **C)** P=V2/R **D)** P=VI

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q182) The gravitational potential energy of an object at a height is the work done to lift it from the ground to that height _____.

A) With constant velocity **B)** Along a curved path **C)** Against gravity **D)** In the direction of gravity

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q183) The main reason for twinkling of stars is _____.

A) Presence of clouds **B)** Nuclear reactions inside stars **C)** Atmospheric refraction varying with time **D)** Scattering of star light

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q184) Which statement correctly describes the universal law of gravitation?

A) The gravitational force increases as the distance between objects increases.

B) The gravitational force increases with the product of the masses and decreases with the square of the distance between them.

C) The gravitational force decreases when the masses of the objects increase.

D) The gravitational force depends only on the distance between the objects, not their masses.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q185) According to Bohr's model, where are electrons found in an atom?

A) Moving in fixed circular orbits around the nucleus **B)** Fixed positions inside the nucleus **C)** Randomly scattered inside the atom **D)** Resting outside the atom in free space

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q186) The weight of a body on the moon is 3 N. What will be the weight of the body on Earth?

A) 3 N **B)** 18 N **C)** 0.5 N **D)** 2 N

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q187) A sound source has a frequency of 50 Hz. How many vibrations does it complete in one minute?

A) 5000 **B)** 50 **C)** 600 **D)** 3000

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q188) The motion of the moon around the earth requires a centripetal force, which is provided by _____.

A) The thrust of the sun **B)** The inertia of the moon **C)** The force of attraction of the earth **D)** The internal forces of the moon

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q189) An object's mass remains constant everywhere (on the Earth or the Moon), whereas its weight changes because _____.

A) The SI unit of weight is Newtons, while mass is kilograms

B) Weight depends on the local value of g, which varies with location **C)** Mass is dependent on inertia, but weight is not

D) Weight is a scalar quantity, while mass is vector

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q190) An electric oven has a power of 1200 W when it is operated at 100 V. How much will be the current flowing through the oven?

A) 12 A **B)** 8 A **C)** 1.2 A **D)** 0.08 A

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q191) In a velocity–time graph, the graph is a straight horizontal line at 10 m/s for 5 seconds. Then the distance travelled is _____.

A) 2 m **B)** 50 m **C)** 10 m **D)** 5 m

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q192) The gravitational force between two masses increases when _____.

A) The product of their masses increases **B)** They are kept in water **C)** Their masses decrease

D) The distance between them increases

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q193) Which of the following statements correctly describes the difference between mass and weight?

A) Mass changes from one planet to another, but weight stays the same. **B)** Both mass and weight change with location.

C) Mass stays the same everywhere, but weight changes with location.

D) Weight and mass both remain constant everywhere.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q194) An object has a mass of 12 kg and moves with a constant speed of 8 m/s. How much kinetic energy does it have?

A) 256 J **B)** 768 J **C)** 512 J **D)** 384 J

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q195) Which of the following is the best example of uniform circular motion?

A) A roller coaster on a loop **B)** A racing car on a curved track slowing down

C) Moon revolving around the Earth in a circular orbit **D)** A yo-yo being swung in a vertical circle at changing speed

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q196) Which of the following is represented by a circle with the letter 'V' in a circuit diagram?

A) Voltage regulator **B)** Variable resistance **C)** Vacuum diode **D)** Voltmeter

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q197) A constant force of 21 N moves a body through 3 m in its own direction. Work done is _____.

A) 24 J **B)** 63 J **C)** 18 J **D)** 7 J

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q198) Why is tungsten used almost exclusively for making filaments of electric bulbs?

A) It has low electrical resistance. **B)** It readily oxidizes at high temperatures. **C)** It does not allow heat to pass.

D) It has a very high melting point.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q199) An object with a mass of 15 kg is placed 8 m above the ground. How much potential energy does the object have? (Using g = 9.8 m/s^2)

A) 1500 J **B)** 784 J **C)** 1176 J **D)** 980 J

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q200) A porter lifts a 10 kg luggage from the ground and places it on his head at a height of 2.5 m. What is the work done by the porter on the luggage? (Using g =10 m/s^2)

A) 25 J **B)** 500 J **C)** 100 J **D)** 250 J

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q201) Which of the following is the correct relation between the object distance (u), image distance (v) and focal length (f) of a convex lens?

A) (1/v) - (1/u) = 1/f **B)** v + u = f **C)** v - u = f **D)** (1/v) x (1/ u) = 1/f

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q202) Which characteristic defines non-uniform motion?

A) Covering unequal distances in equal intervals of time **B)** Equal displacement in equal time intervals **C)** Constant speed

D) Zero acceleration

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q203) A cyclist riding at constant speed on a circular track is an example of which type of motion?

A) Non-uniform motion **B)** Linear motion **C)** Uniform circular motion **D)** Oscillatory motion

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q204) A body undergoes a displacement of 50 m under the action of a constant force of 5 N in the same direction as that of the force. What will be the work done under the action of this force?

A) + 25 J **B)** - 250 J **C)** - 25 J **D)** + 250 J

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q205) What does the given circuit symbol represent?



A) Variable resistance or rheostat **B)** A wire **C)** A battery **D)** Switch in the closed position

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q206) When we observe a star near the horizon, it appears to be slightly higher in the sky than it really is. What is the reason for this?

A) The star moves rapidly and changes its position. **B)** The star becomes brighter, making it seem higher.

C) The Earths rotation shifts the star upward.

D) The atmosphere bends the starlight due to gradual changes in refractive index.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q207) If a conductor has a potential difference of 1 V across its ends and a current of 1 A flows through it, what is the resistance of the conductor?

A) 0.5 Ω **B)** 2 Ω **C)** 10 Ω **D)** 1 Ω

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q208) What is the work done on an object if its displacement is zero, even when a force is applied?

A) Zero **B)** Infinite **C)** Negative **D)** Positive

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q209) What is the audible frequency range for human beings?

A) 200 Hz to 200,000 Hz **B)** 20 kHz to 200 kHz **C)** 2 Hz to 200 Hz **D)** 20 Hz to 20,000 Hz

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q210) SONAR technology uses ultrasound to _____.

A) Measure electric current **B)** Produce high temperatures **C)** Locate underwater objects **D)** Measure atmospheric pressure

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Chemistry (Q-110)

Q211) Which of the following pairs are correctly matched as structural isomers?

A) Pentane – 2-methylbutane

B) Butane – 2-methylpropane

C) Benzene – cyclohexene

A) A and B only **B)** B and C only **C)** A and C only **D)** A, B and C

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q212) Identify the product formed when copper metal is exposed to dilute hydrochloric acid.

A) Salt and chlorine gas **B)** Salt and hydrogen gas **C)** No reaction **D)** Only Salt

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q213) Read the following three statements and choose the correct option:

Statement I: Formula unit mass is applicable only to ionic compounds, not covalent ones.
Statement II: Formula unit mass is the sum of the atomic masses of all atoms in the empirical formula of an ionic compound.
Statement III: The formula unit mass of sodium chloride (NaCl) is 58.5 u.

A) Only I and II are correct. **B)** All I, II and III are correct. **C)** Only I and III are correct. **D)** Only II and III are correct.

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q214) Select the correct option regarding the given statements.

Statement 1: Metals and non-metals react by sharing electrons.
Statement 2: The bond formed between them is covalent.

A) Both statements are true. **B)** Statement 1 is true; Statement 2 is false. **C)** Statement 1 is false; Statement 2 is true.

D) Both statements are false.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q215) The solubility of **gases** in liquids generally:

A) Is unaffected by temperature **B)** Increases with increase in temperature **C)** Decreases with increase in temperature

D) First increases then decreases with increase in temperature

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q216) What is the concentration of H^+ ions in pure water at 25°C ?

A) 1×10^{-5} M **B)** 1×10^{-1} M **C)** 1×10^{-4} M **D)** 1×10^{-7} M

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q217) If two isotopes of an element 'X' are X-35 and X-37, the average atomic mass of chlorine (if X-35 = 25% and X-37 = 75%) will be:

A) 35.5 u **B)** 36.0 u **C)** 34.5 u **D)** 36.5 u

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q218) In the substitution reaction of methane (CH_4) with chlorine (Cl_2) under UV light, what is the first product formed?

A) Methyl chloride (CH3Cl) **B)** Chloromethane (CH3Cl) **C)** Dichloromethane (CH2Cl2) **D)** Carbon tetrachloride (CCl4)

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q219) Consider the following statements regarding the size of an atom.

Statement 1: The size of an atom is approximately in the range of 0.1 nm to 0.5 nm.

Statement 2: Atomic size increases as we move from left to right in a period of the periodic table.

Statement 3: The radius of a hydrogen atom is about 10^{-10} meters.

Which of the above statements is/are correct?

A) Only 2 and 3 **B)** Only 1 and 2 **C)** Only 1, 2 and 3 **D)** Only 1 and 3

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q220) The compound with molecular formula C_2H_4O can belong to which of the following functional group, and what is its IUPAC name?

A) Aldehyde, ethanol **B)** Aldehyde, ethanal **C)** Ether, ethane **D)** Alcohol, ethanol

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q221) An element X has two isotopes: ${}_6X^{12}$ and ${}_6X^{14}$. Both are electrically neutral and chemically identical, but one of them is unstable and radioactive. Which of the following statements best explains why ${}_6X^{14}$ is radioactive, even though both isotopes have the same number of protons and electrons?

A) Because ${}_6X^{14}$ has a smaller nucleus that cannot hold neutrons tightly, resulting in spontaneous decay

B) Because ${}_6X^{14}$ has excess neutrons, causing imbalance in nuclear forces

C) Because ${}_6X^{14}$ has equal numbers of protons and neutrons, making it energetically unstable

D) Because ${}_6X^{14}$ has more electrons than protons, leading to instability in electron distribution

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q222) The M shell can take a maximum of ____ electrons, but in Ca it has _____ electrons. [Atomic Number of Ca = 20]

A) 18, 10 **B)** 18, 8 **C)** 8, 8 **D)** 8, 10

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q223) Select the correct option regarding the given statements.

Statement 1: The compound $CH_3–CH_2–CHO$ is propanone.
Statement 2: Aldehydes are named using the suffix '–al'.

A) Statement 1 is false but Statement 2 is true.

B) Both statements are true but Statement 2 is not the correct explanation of Statement 1.

C) Statement 1 is true but Statement 2 is false.

D) Both statements are true and Statement 2 is the correct explanation of Statement 1.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q224) The resistivity of an alloy is generally higher than its constituent metals, and this property makes alloys suitable for _____.

A) Connecting wires (e.g., copper transmission lines) **B)** Filaments of electric bulbs (e.g., tungsten)

C) Heating elements (e.g., nichrome) **D)** Circuit fuses (e.g., lead/tin alloy)

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q225) During the extraction of metals from their ores, the removal of gangue is an important preliminary step. Which of the following statements most accurately explains why different concentration methods (such as hydraulic washing, froth flotation and magnetic separation) are used for different ores?

A) Because gangue particles are chemically reactive and must be removed by oxidation before extraction

B) Because all ores have identical physical properties but different chemical compositions

C) Because each method exploits specific differences in physical and chemical properties between the ore and the gangue

D) Because the choice of concentration method depends on the melting point of the ore and gangue

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q226) Which statement is **INCORRECT** about isobars?

A) Their number of neutrons is different. **B)** They have the same mass number. **C)** They have the same number of protons.

D) They have different chemical properties.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q227) Which of the following methods is commonly used to extract metals low in the activity series, such as mercury and copper, from their ores?

A) Displacement by carbon monoxide gas **B)** Reduction with aluminium powder **C)** Roasting **D)** Electrolysis of molten ore

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q228) If the pH of a solution is 3, what is the concentration of hydrogen ions in the solution?

A) $3\times10-2M$ **B)** $1\times10-3$ M **C)** $3\times10-3$ M **D)** $1\times10-2$ M

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q229) Which of the following are properties of ionic compounds?

A. High melting point
B. Soluble in non-polar solvents
C. Conduct electricity in solid state
D. Brittle in nature

A) Only (C) and (B) **B)** Only (A) and (B) **C)** Only (C) and (D) **D)** Only (A) and (D)

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q230) In the process of electrolytic refining of copper, which of the following is correctly matched?

A) Cathode – Impure copper **B)** Electrolyte – Acidified Copper sulphate solution **C)** Cathode Mud – Pure copper crystals

D) Anode – Pure copper

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q231) Four test tubes contain solutions of $CuSO_4$, $MgSO_4$, $ZnSO_4$, and $AgNO_3$. A student adds zinc metal to each test tube. In which test tube(s) will a reaction occur?

A) With all solutions **B)** $CuSO_4$ only **C)** $CuSO_4$ and $AgNO_3$ only **D)** $ZnSO_4$ only

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q232) In the electrolytic refining of copper, which of the following acts as the **anode**?

A) Pure copper plate **B)** Copper sulphate solution **C)** Carbon electrode **D)** Impure copper block

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q233) Which condition is required for methane to undergo a substitution reaction with chlorine?

A) Acidic medium **B)** Heat **C)** Electric current **D)** Sunlight

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q234) Dhritiman tries to extract magnesium from magnesium oxide using carbon. The attempt fails. What is the reason?

A) Carbon reacts violently with oxygen. **B)** The reaction produces carbon monoxide only.

C) Magnesium forms a volatile compound with carbon. **D)** Carbon is less reactive than magnesium.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q235) Which of the following is the correct formula for the compound calcium phosphate?

A) Ca4(PO4)3 **B)** Ca3(PO4)2 **C)** CaPO4 **D)** Ca2(PO4)3

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q236) In chemical industries, bleaching powder acts mainly as:

A) A reducing agent **B)** A neutralising agent **C)** An oxidising agent **D)** A catalyst

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q237) Ethanol and ethanoic acid react in the presence of concentrated sulphuric acid to form a sweet-smelling compound. Which of the following correctly identifies the reaction type and the role of concentrated H_2SO_4?

A) It is a neutralization reaction; H_2SO_4 acts as an acid to neutralize the base.

B) It is an oxidation reaction; H_2SO_4 acts as an oxidizing agent.

C) It is an esterification reaction; H_2SO_4 acts as a catalyst and a dehydrating agent.

D) It is a substitution reaction; H_2SO_4 acts as a dehydrating agent to remove water molecules.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q238) Which of the following statements is correct regarding molecules of an element?

A) Molecules of an element always contain two or more different types of atoms.

B) All elements exist as monoatomic molecules in their natural state.

C) A molecule of an element consists of only one type of atom bonded together.

D) Elements do not form molecules, only compounds do.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q239) Which of the following correctly explains why toothpaste is basic in nature?

A) To neutralise the acid formed in the mouth after eating food **B)** To provide an acidic medium for enzyme action in saliva

C) To react with water and release hydrogen ions **D)** To lower the pH of saliva and prevent cavities

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q240) Which of the following is a ring compound with the molecular formula C_6H_{12}?

A) Butane **B)** Benzene **C)** Cyclohexane **D)** Hexane

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q241) Burning of a saturated hydrocarbon will result in which type of flame?

A) Clean flame **B)** Sooty flame **C)** Yellow flame **D)** Green flame

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q242) 1. A precipitation reaction occurs when two aqueous solutions react to form an insoluble solid.
2. The solid formed is called a precipitate.
3. All precipitation reactions involve acids and bases only.

Which of the above statements are correct?

A) 1, 2, and 3 **B)** 2 and 3 only **C)** 1 and 3 only **D)** 1 and 2 only

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q243) Read the following statements carefully and choose the correct option.

Statement 1: Members of a homologous series differ by a $–CH_2–$ group.
Statement 2: Each successive member of a homologous series differs from the previous one by 12 u in molecular mass.

A) Both statements are false. **B)** Both statements are true but Statement 2 is not the correct explanation of Statement 1.

C) Both statements are true and Statement 2 is the correct explanation of Statement 1.

D) Statement 1 is true but Statement 2 is false.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q244) Which of the following statements about natural acid-base indicators is correct?

A) Red litmus turns blue in acids. **B)** Turmeric turns red in basic solution.

C) Methyl orange is a natural indicator extracted from turmeric. **D)** China rose (Hibiscus) extract turns green in bases.

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q245) Which of the following statements is **FALSE** about mass number?

A) It represents total nucleons in the atom. **B)** It is always a whole number.

C) It is equal to twice the atomic number for all elements. **D)** It may differ for isotopes of the same element.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q246) Addition reactions are characteristic of which type of hydrocarbons?

A) Both alkynes and alkenes **B)** Only alkynes **C)** Only alkenes **D)** Only alkanes

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q247) During the formation of coal from plant material, a sequence of chemical and physical changes occurs under high pressure and temperature over millions of years. Which of the following correctly represents the progressive increase in carbon content during the formation of coal?

A) Peat → Anthracite → Bituminous → Lignite **B)** Peat → Bituminous → Lignite → Anthracite

C) Peat → Lignite → Bituminous → Anthracite **D)** Lignite → Peat → Anthracite → Bituminous

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q248) Which of the following pairs of compounds contain the same total number of atoms per molecule?

A) NH_3 and CH_4 **B)** H_2SO_4 and N_2O_5 **C)** CO_2 and H_2O_2 **D)** C_2H_6O and PCl_3

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q249) The difference in electrical conductivity between diamond and graphite arises because:

A) Graphite has delocalized electrons; diamond does not **B)** Diamond has metallic bonding, graphite does not

C) Diamond has tightly packed ions **D)** Both have free electrons, but in different numbers

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q250) Which of these is an example of a solid sol?

A) Coloured gem **B)** Jelly **C)** Cheese **D)** Smoke

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q251) Which of the following statements are correct?

1. Colloids appear homogeneous because their particles are very small.
2. Colloids are truly homogeneous at the molecular level.
3. Colloidal particles scatter light.
4. Milk is a colloidal solution.

A) 2 and 4 only **B)** 1, 2, 3 and 4 **C)** 1 and 2 only **D)** 1, 3 and 4 only

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q252) Metals in the middle of the activity series, such as iron and zinc, are generally extracted by:

A) Heating alone without any reducing agent **B)** Reduction of their oxides with carbon or carbon monoxide

C) Electrolysis of molten salts **D)** Displacement from aqueous solution

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q253) Which of the following are produced when a metal oxide reacts with an acid?

A) Salt and carbon dioxide **B)** Salt and hydrogen gas **C)** Salt and water **D)** Salt and oxygen gas

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q254) Read the given statements carefully and select the correct option.

Statement-I: Most carbon compounds are poor conductors of electricity.
Statement-II: Carbon compounds have low melting and boiling points as compared to ionic compounds.

A) Both the statements are false. **B)** Statement-I is false but statement-II is true.

C) Statement-I is true but statement-II is false. **D)** Both the statements are true.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q255) Why is electrolysis used for the extraction of metals high up in the reactivity series, such as sodium and aluminum?

A) Because these metals exist in native state **B)** Because these metals have low melting points

C) Because these metals cannot be reduced by carbon or hydrogen **D)** Because these metals form volatile oxides

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q256) The molecular mass of a compound is '60'. If it contains 40% carbon, 6.7% hydrogen, and 53.3% oxygen, what is the molecular formula?

A) $C_2H_4O_2$ **B)** $C_3H_6O_3$ **C)** CH_2O **D)** $C_4H_8O_4$

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q257) When a small piece of sodium metal is added to chlorine gas under controlled conditions, a vigorous reaction occurs producing sodium chloride. Which of the following statements best explains the reason for the stability of the product formed?

A) Sodium donates one electron to chlorine, forming oppositely charged ions held by strong electrostatic forces.

B) Both sodium and chlorine attain stability by forming a metallic lattice with delocalized electrons.

C) Sodium and chlorine share one electron each, forming a stable covalent bond.

D) Sodium gains one electron from chlorine, forming a neutral molecule through mutual sharing.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q258) What will be the pH of a 0.01 M HCl solution?

A) 8 **B)** 18 **C)** 2 **D)** 12

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q259) A clean strip of calcium metal is added to cold water and the reaction is observed. Compared to sodium and magnesium, which of the following statements correctly describes calcium's reactivity and the nature of products formed?

A) Calcium reacts less vigorously than sodium but more than magnesium, forming calcium hydroxide and hydrogen gas.

B) Calcium reacts explosively like sodium, forming calcium hydroxide and hydrogen gas.

C) Calcium reacts slowly with cold water, producing calcium oxide and oxygen gas.

D) Calcium does not react with cold water but only with steam, forming calcium oxide and hydrogen gas.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q260) In the chemical equation $xPb(NO_3)_2(s) \rightarrow yPbO(s) + zNO_2(g) + wO_2(g)$, what are the values of x, y, z and w?

A) x = 2 ; y = 4 ; z = 2 ; w = 1 **B)** x = 1 ; y = 2 ; z = 3 ; w = 2 **C)** x = 2 ; y = 1 ; z = 4 ; w = 2 **D)** x = 2 ; y = 2 ; z = 4 ; w = 1

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q261) The atomic number of oxygen is 8. What does this mean?

A) Oxygen has 8 protons and 8 electrons **B)** Oxygen has 8 atoms in each molecule **C)** Oxygen has 8 neutrons in its nucleus

D) Oxygen has 8 protons in its orbit

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q262) Two beakers contain sugar solution of different concentrations. A semipermeable membrane separates them. After some time, the level of solution in one beaker rises while in the other, it falls. Which property of solutions does this experiment best demonstrate, and why?

A) Sedimentation – because sugar crystals settle at the bottom due to gravity

B) Diffusion – because sugar molecules move freely across the membrane until equilibrium is reached

C) Evaporation – because water molecules escape from the surface of the solution causing a level difference

D) Osmosis – because solvent molecules move from the dilute solution to the concentrated solution through the semipermeable membrane

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q263) If both solute and solvent masses are doubled, the concentration (mass %) of the solution will:

A) Remain the same **B)** Double **C)** Become half **D)** Increase by 25%

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q264) Which of the following salts, when dissolved in water, will produce a neutral solution?

A) Sodium chloride (NaCl) **B)** Sodium acetate (CH3COONa) **C)** Ammonium chloride (NH4Cl) **D)** Potassium hydroxide (KOH)

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q265) Which of the following best defines reduction in a redox reaction?

A) Loss of electrons only **B)** Gain of oxygen or loss of hydrogen **C)** Gain of hydrogen or loss of oxygen

D) Gain of both oxygen and electrons

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q266) Which of the following is NOT a correct use or property of sodium hydrogen carbonate, $NaHCO_3$?

A) It is used in fire extinguishers. **B)** It is used in baking as a leavening agent. **C)** It is used as a strong acid.

D) It is used in antacids.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q267) Which of the following is NOT a reason for carbon's versatility?

A) Catenation **B)** Ability to form single, double and triple bonds **C)** Tetravalency **D)** Formation of ionic bonds

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q268) Which of the following pairs must represent the same element?

A) Same mass number, different atomic numbers **B)** Same number of neutrons, different atomic numbers

C) Different number of protons, same number of neutrons **D)** Same atomic number, different mass numbers

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q269) What is the formula unit mass of ammonium sulphate $(NH_4)_2SO_4$?

A) 146 u **B)** 114 u **C)** 132 u **D)** 98 u

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q270) Which of the following compounds obeys the Law of Constant Proportions?

A) Brass **B)** Air **C)** Water **D)** Soil

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q271) What is the main advantage of using olfactory indicators?

A) They give exact pH values. **B)** They change colour sharply. **C)** They can be used for all solutions.

D) They are useful for visually impaired individuals.

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q272) Read the given statements carefully and select the correct option.

Statement I: The phenomenon of change of liquid into vapours at any temperature below its boiling point is called evaporation.

Statement II: The rate of evaporation increases with an increase of surface area.

A) Statement I is false but statement II is true. **B)** Statement I is true but statement II is false.

C) Both statements I and II are true. **D)** Both statements I and II are false.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q273) What is produced when an acid reacts with a metal hydrogen carbonate?

A) Salt, oxygen and water **B)** Salt, carbon dioxide and water **C)** Salt, hydrogen gas and water **D)** Salt, ammonia and water

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q274) Which of the following statements correctly differentiates a compound from a mixture?

A) Compounds show variable composition, mixtures do not. **B)** Mixtures are formed through chemical reactions.

C) Mixtures can be separated by physical means, but compounds cannot.

D) Compounds retain the properties of their constituents.

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q275) In a homogeneous mixture, the size of particles of solute is typically:

A) Between 1 nm and 1000 nm **B)** Less than 1 nm **C)** Larger than 1 μm **D)** Visible to the naked eye

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q276) A student heats 10 g of calcium carbonate in a closed container. It decomposes into 5.6 g of calcium oxide and carbon dioxide gas. What will be the mass of carbon dioxide produced?

A) 5.0 g **B)** 15.6 g **C)** 4.4 g **D)** 10.0 g

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q277) If an element forms an ion with a charge of −2, what does this indicate about its valency?

A) The element has lost 2 electrons. **B)** The element has gained 2 electrons. **C)** The element has 2 electrons in the nucleus.

D) The element has gained 2 protons.

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q278) Ethanoic acid reacts with sodium carbonate to produce:

A) Sodium hydroxide and CO_2 **B)** Sodium acetate, hydrogen, and water **C)** Sodium acetate and carbon monoxide

D) Sodium acetate, carbon dioxide, and water

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q279) Which of the following metals forms an amphoteric oxide when reacting with air?

A) Calcium **B)** Sodium **C)** Aluminium **D)** Potassium

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q280) When potassium reacts with water, we observe:

A) No visible change **B)** Formation of a greenish compound **C)** Gentle bubbling and slow reaction

D) Bright flame and rapid reaction

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q281) Which of the following is the method used for refining metals such as copper, zinc and silver?

A) Distillation **B)** Extraction **C)** Electrolytic refining **D)** Smelting

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q282) Which calculation correctly determines the formula unit mass of $CaCl_2$?

A) Atomic mass of Ca + (atomic mass of Cl ÷ 2) = 40 + 17.75 = 57.75 u

B) Atomic mass of Ca + (2 × atomic mass of Cl) = 40 + 71 = 111 u

C) Atomic mass of Ca + atomic mass of Cl = 40 + 35.5 = 75.5 u

D) Atomic mass of Ca × atomic mass of Cl = 40 × 35.5 = 1420 u

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q283) If the molecular mass of an alkane is **44**, then its molecular formula is:

A) C_2H_6 **B)** C_4H_{10} **C)** C_3H_8 **D)** CH_4

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q284) Which of the following pairs are isotopes?

A) ^{1}H and ^{2}H **B)** ^{16}O and ^{16}F **C)** ^{12}C and ^{14}N **D)** ^{40}Ca and ^{40}Ar

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q285) Which of the following is the product formed when ethene (C_2H_4) reacts with hydrogen (H_2) in the presence of a catalyst nickel?

A) Propene (C3H6) **B)** Ethyne (C2H2) **C)** Propane (C3H8) **D)** Ethane (C2H6)

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q286) Which of the following is a characteristic feature of all organic compounds?

A) They always form ionic bonds **B)** They always contain carbon **C)** They always contain oxygen

D) They always contain nitrogen

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q287) Which of the following statement about a solution is correct?

A) A solution is a homogeneous mixture in which the solute particles are not visible to the naked eye.

B) A solution is a heterogeneous mixture of two or more substances. **C)** The solute in a solution settles down on standing.

D) The solute in a solution is always present in a greater amount than the solvent.

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q288) The burning of carbon to form carbon dioxide is an example of a/an:

A) Endothermic reaction **B)** Photochemical reaction **C)** Exothermic reaction **D)** Reversible reaction

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q289) The versatile nature of carbon is responsible for the formation of which of the following types of compounds?

A) Organic compounds **B)** Inorganic compounds **C)** Metallic compounds **D)** Ionic compounds

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q290) Which of the following correctly represents a diatomic element?

A) H_2 **B)** He **C)** H **D)** CO

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q291) The formation of micelles in soap solution occurs due to:

A) hydrophobic and hydrophilic interactions in water **B)** electrostatic attraction between ions

C) hydrogen bonding with water molecules **D)** dispersion forces only

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q292) What is the main difference between isobars and isotopes?

A) Isobars have the same mass number but different atomic numbers

B) Isobars have the same number of electrons; isotopes do not

C) Isobars have the same atomic number; isotopes have different atomic numbers

D) Isobars are atoms of the same element; isotopes are of different elements

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q293) Along with phosphorus and potassium, fertilisers mainly supply which of the following to crops?

A) Zinc only **B)** Calcium only **C)** Nitrogen **D)** Magnesium only

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q294) The ratio of the unit mass of a molecule of nitrogen (N_2) to that of an oxygen molecule (O_2) is approximately:

A) 8 : 7 **B)** 7 : 8 **C)** 14 : 32 **D)** 28 : 16

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q295) Which of the following is the best method to prevent corrosion of iron in coastal areas?

A) Immersing in cold water **B)** Painting with oil-based paints **C)** Galvanisation **D)** Heating the iron periodically

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q296) Which of the following is a molecule formed by two atoms of the same element?

A) H_2 **B)** H_2O **C)** CO **D)** O_3

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q297) Match the elements in Column I with their correct electron distribution in Column II according to the Bohr–Bury rules.

Element	Electron Distribution
A. Sodium (Na)	1. 2, 8, 2
B. Magnesium (Mg)	2. 2, 8, 1
C. Aluminium (Al)	3. 2, 8, 3
D. Calcium (Ca)	4. 2, 8, 8, 2

A) A–2, B–1, C–3, D–4 **B)** A–2, B–3, C–1, D–4 **C)** A–2, B–3, C–4, D–1 **D)** A–1, B–3, C–2, D–4

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q298) Read the given statements carefully. Select which statement(s) is/are correct.

Statement-I: There are some substances whose odour changes in acidic or basic media. These are called olfactory indicators.

Statement-II: Vanilla extract and onion can be used as olfactory indicators.

A) Both statements I and II are true. **B)** Statement-I is true but statement-II is false. **C)** Both statements I and II are false.

D) Statement-I is false but statement-II is true.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q299) Which statement is true about mass number?

A) It is always twice the atomic number **B)** It is the sum of protons and neutrons in an atom

C) It is equal to the number of electrons in an atom **D)** It is the total number of protons and electrons in an atom

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q300) Which of the following isotopes is used in the treatment of cancer?

A) Isotope of uranium **B)** Isotope of iodine **C)** Isotope of cobalt **D)** Isotope of carbon

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q301) Which of the following correctly represents the unit mass of a molecule of water (H_2O)?

A) 18 u **B)** 10 u **C)** 16 u **D)** 20 u

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q302) Which of the following non-metals is a good conductor of electricity in the solid state due to the presence of delocalized electrons within its crystal lattice?

A) Phosphorus **B)** Iodine **C)** Graphite **D)** Sulphur

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q303) In a chemical reaction, 5.6 g of sodium reacts with 8.0 g of chlorine to form sodium chloride. What is the mass of sodium chloride formed according to the Law of Conservation of Mass?

A) 8.0 g **B)** 12.4 g **C)** 13.6 g **D)** 14.4 g

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q304) Which of the following is NOT a combination reaction?

A) $2Cu + O_2 \xrightarrow{Heat} 2CuO$ **B)** $CuO + H_2 \xrightarrow{Heat} Cu + H_2O$ **C)** $NH_3 + HCl \rightarrow NH_4Cl$

D) $CaO + H_2O \rightarrow Ca(OH)_2$

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q305) Which of the following statements is true with respect to the reaction of a metal 'A' with metal 'B' of its solution, in which metal 'A' displaces metal 'B'?

A) 'A' is more reactive than 'B'. **B)** 'B' is more reactive than 'A'. **C)** Both 'A' and 'B' have the same reactivity.

D) The reaction cannot take place at all.

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q306) Which of the following statements correctly explains the position of sodium (Na) in the activity series of metals?

A) Sodium is less reactive than copper and cannot displace it from salt solution.

B) Sodium is less reactive than hydrogen and hence placed below it.

C) Sodium is more reactive than hydrogen and displaces it from acid.

D) Sodium does not react with water, hence placed at the bottom of the series.

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q307) If A displaces B, and C displaces A, then the order of reactivity is:

A) B > A > C **B)** C > A > B **C)** C > B > A **D)** A > B > C

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q308) Which property do isobars share?

A) Same chemical properties **B)** Same mass number **C)** Same number of protons **D)** Same atomic number

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q309) Which of the following is correct for substitution reaction?

A) A reaction where hydrogen is added to a compound.

B) A reaction where a compound breaks down into simpler substances.

C) A reaction where atoms or groups in a molecule are replaced by another atom or group.

D) A reaction where two compounds combine to form one.

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q310) The symbol 'B' represents which element?

A) Bromine **B)** Barium **C)** Beryllium **D)** Boron

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q311) Which of the following is an example of a double displacement reaction?

A) Fe + $CuSO_4$ → $FeSO_4$ + Cu **B)** $2H_2 + O_2 \rightarrow 2H_2O$ **C)** $CaCO_3 \rightarrow CaO + CO_2$ **D)** $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q312) If 9 g of water contains 1 g of hydrogen and 8 g of oxygen, then 36 g of water will contain:

A) 4 g of hydrogen and 32 g of oxygen **B)** 3 g of hydrogen and 32 g of oxygen **C)** 4 g of hydrogen and 24 g of oxygen **D)** 3 g of hydrogen and 24 g of oxygen

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q313) Isotopes are atoms of the same element that have:

A) the same atomic number but different mass numbers **B)** different number of protons **C)** different chemical properties **D)** the same mass number but different atomic numbers

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q314) Which of the following pairs of element names and symbols is correctly matched?

A) Mercury – Me **B)** Lead – Ld **C)** Silver – S **D)** Tin – Sn

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q315) Which of the following observations best confirms that a neutralisation reaction has occurred?

A) The temperature of the solution drops drastically. **B)** The solution changes from blue to red. **C)** The pH of the solution becomes close to 7. **D)** A gas with a pungent smell is evolved.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q316) A scientist observed that an iron railing near the sea corroded faster than the same type of railing in a dry city. What could the main reason be?

A) Sea air contains salt and moisture, which accelerates rusting. **B)** The sea air is more humid, diluting the rusting process. **C)** The iron near the sea is exposed to more sunlight. **D)** The city air has less oxygen, preventing rust.

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q317) Which of the following is an example of a double displacement reaction?

A) Na2O + H2O→ 2 NaOH **B)** $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$ **C)** $2Mg + O_2 \rightarrow 2MgO$ **D)** $H_2 + Cl_2 \rightarrow 2HCl$

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q318) Which method involves coating iron with a layer of zinc to protect it from rusting?

A) Galvanisation **B)** Electroplating **C)** Tinning **D)** Painting

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q319) What is the formula unit mass of sodium carbonate (Na_2CO_3)?
(Atomic masses: Na = 23 u, C = 12 u, O = 16 u)

A) 98 u **B)** 84 u **C)** 62 u **D)** 106 u

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q320) A student mixes sand, salt and water in a beaker and stirs well. After some time, what will be observed?

A) Sand and salt both remain undissolved, forming two distinct layers.

B) All components dissolve completely, forming a clear solution.

C) Sand settles at the bottom, while salt dissolves in water.

D) Sand remains permanently suspended in water, making the mixture cloudy.

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

SelectionWay

Biology (Q-105)

Q321) Choose a term to complete the analogy.

Xylem Parenchyma : Food Storage :: Xylem Fibres : _____

A) Transporting minerals **B)** Supportive function **C)** Being tubular structures **D)** Transporting water vertically

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q322) Choose a term to complete the analogy.

Columnar Epithelium : Absorption :: Cuboidal Epithelium : _____

A) Mechanical support **B)** Pushing mucus **C)** Forming a delicate lining **D)** Preventing wear and tear

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q323) Which of the following statements correctly explains the working principle of dialysis in an artificial kidney?

A) Dialysis removes wastes from blood by diffusion across a semi-permeable membrane.

B) Dialysis removes nitrogenous wastes from blood through active transport.

C) Dialysis uses high pressure to force wastes out of the blood into the dialysing fluid.

D) Dialysis reabsorbs essential nutrients and salts like a normal kidney.

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q324) When observing onion peel and cheek cells under a microscope, which part appears lightly stained and contains many organelles?

A) Nucleus **B)** Cytoplasm **C)** Cell membrane only **D)** Cell wall only

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q325) Which of the following components is absent in lymph but present in blood?

A) Platelets **B)** Plasma **C)** Water **D)** White blood cells (WBCs)

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q326) Which of the following best distinguishes a food web from a food chain?

A) A food web allows backward energy flow; a food chain is unidirectional

B) A food web shows multiple trophic levels; a food chain shows one level

C) A food chain always has decomposers; a food web does not

D) A food web transfers all energy; a food chain loses energy

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q327) Which cropping pattern maintains soil fertility through planned crop succession?

A) Crop rotation **B)** Monocropping **C)** Mixed cropping **D)** Relay cropping

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q328) Choose a term to complete the analogy.

Chlorenchyma : Photosynthesis :: Aerenchyma : _____

A) Food storage **B)** Water absorption **C)** Buoyancy **D)** Mechanical strength

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q329) What specialisation is commonly found in the epidermal cells of plant roots to aid in their primary function?

A) Irregularly thickened corners **B)** Large air cavities (aerenchyma) **C)** A thick, waxy coating of cutin

D) Long, hair-like extensions

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q330) Which of the following correctly explains the adaptation of desert plants in photosynthesis?

A) Light energy is captured at night and converted into starch for daytime use.

B) Oxygen is absorbed at night while carbon dioxide is released during the day.

C) Carbon dioxide uptake occurs at night and carbohydrates are synthesized during the day.

D) Photosynthesis is restricted to night-time only to minimise water loss.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q331) Plants get macro-nutrients mainly from the _____, and micro-nutrients in small quantities from the _____.

A) soil ; soil **B)** water ; sand **C)** sunlight ; minerals **D)** air ; water

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q332) Which of the following movements is controlled by unstriated and involuntary smooth muscle tissue?

A) Moving the legs to walk **B)** Rhythmic contraction of the heart **C)** Movement of food in the alimentary canal

D) Waving the hand

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q333) Which description correctly characterises parenchyma cells in the pith of a mature stem?

A) Large, loosely arranged living cells for storage. **B)** Tightly packed dead cells that aid in support and conduction.

C) Thick-walled lignified cells for mechanical strength. **D)** Sclerenchyma fibers forming a protective cylinder.

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q334) Which of the following statements best explains the functional advantage of the pollen tube formation in the process of fertilisation in flowering plants?

A) It allows the pollen grain to attach firmly to the stigma for self-pollination to occur efficiently

B) It helps the pollen grain to release its stored nutrients for the ovule to develop into a seed

C) It ensures direct transfer of the male germ-cell to the ovary, preventing wastage of pollen grains

D) It enables the ovary to release egg cells toward the pollen grain for fusion to take place

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q335) _____ is a process in which persistent, non-degradable chemicals become progressively more concentrated at each higher trophic level in a food chain.

A) Biological magnification **B)** Ecological succession **C)** Bioremediation **D)** Biodegradation

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q336) Which type of epithelial tissue is primarily specialized for absorption in the human intestine?

A) Simple cuboidal epithelium **B)** Simple squamous epithelium **C)** Stratified squamous epithelium **D)** Simple columnar epithelium

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q337) What are the tiny blobs present on stick-like structures in *Rhizopus* called?

A) Spores **B)** Hyphae **C)** Roots **D)** Sporangia

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q338) Which property of connective tissue enables it to bind and support different body parts?

A) Presence of cilia **B)** Presence of contractile proteins **C)** Presence of extracellular matrix **D)** Presence of nerve endings

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q339) All of the following functions are performed by different components of a Xylem tissue, EXCEPT:

A) Transporting food from leaves **B)** Transporting water and minerals vertically **C)** Providing mechanical support **D)** Storing food

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q340) Which of the following statements correctly describes the transportation of blood from the heart to the lungs?

A) Deoxygenated blood is pumped from the left ventricle to the lungs.

B) Oxygenated blood is carried from the lungs to the right atrium.

C) Oxygenated blood flows from the left ventricle to the lungs for purification.

D) Deoxygenated blood is pumped from the right ventricle to the lungs.

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q341) The deeply folded inner membrane of mitochondria helps to increase the _____ for ATP generation.

A) Surface area **B)** Cell wall thickness **C)** Colour **D)** Volume

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q342) Implantation of developing embryo occurs in which part of the female reproductive system in human beings?

A) Lining of the uterus **B)** Cervix **C)** Fallopian tubes **D)** Ovary

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q343) Which component of phloem is responsible for the transport of food in plants?

A) Phloem parenchyma **B)** Companion cells **C)** Sieve tubes **D)** Phloem fibers

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q344) Which structural characteristic allows epithelial tissue to function as a protective layer?

A) Cells are connected exclusively by desmosomes. **B)** Cells are long and multinucleated.

C) Cells are tightly packed with minimal intercellular space.

D) Cells are loosely arranged with abundant extracellular matrix.

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q345) Why is the use of condoms during sexual intercourse recommended?

A) Condoms completely eliminate the risk of pregnancy and infections

B) Condoms are used to increase fertility in both males and females

C) Condoms help prevent the transmission of sexually transmitted diseases to some extent

D) Condoms are required by law to maintain the population balance

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q346) In the respiratory tract, columnar epithelium possesses cilia. What is the primary function of these cilia?

A) To move and push mucus forward to clear it **B)** To provide only mechanical support to the tract

C) To secrete substances onto the epithelial surface **D)** To increase the surface area for absorption only

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q347) Why is vegetative propagation useful for agriculture purposes?

A) It increases genetic variation **B)** It requires seeds for every generation **C)** It produces identical plants rapidly

D) It prevents diseases naturally

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q348) How long does an unfertilised egg survive after being released from the ovary?

A) About one day **B)** One week **C)** Two hours **D)** One month

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q349) What causes only the animal cell to burst when both plant and animal cells are placed in pure water?

A) Plant cells can actively pump out water. **B)** Osmosis occurs only in animal cells.

C) Animal cells have a thicker membrane. **D)** Plant cells have a rigid cell wall.

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q350) Refer to the two statements given below and choose the CORRECT option.

Statement A: Nervous tissue is responsible for rapid transmission of electrical impulses in the body.

Statement B: Nervous tissue mainly provides structural and mechanical support to organs.

A) Statement A is correct but B is incorrect. **B)** Both statements A and B are correct.

C) Both statements A and B are incorrect. **D)** Statement A is incorrect but B is correct.

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q351) Which of the following best explains why anaerobic respiration releases less energy than aerobic respiration?

A) It does not require enzymes. **B)** It partially breaks down glucose. **C)** It uses less glucose.

D) It occurs slower than aerobic respiration.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q352) Epithelial tissue primarily functions in:

A) Coordination **B)** Secretion and protection **C)** Support **D)** Muscle contraction

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q353) If producers increase in number and herbivores decrease in number, what is the likely sudden consequence in the food chain?

A) Carnivore population will decrease due to less food. **B)** Producers will decrease due to overgrazing.

C) Carnivore population will increase. **D)** Decomposers population will decline suddenly.

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q354) Which of the following is NOT known to reproduce through fission?

A) Hydra **B)** Paramecium **C)** Leishmania **D)** Amoeba

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q355) What are the two main categories of factors responsible for storage losses in agricultural produce?

A) Biotic and Abiotic **B)** Mechanical and Thermal **C)** Harvesting and Processing **D)** Chemical and Electrical

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q356) Read the given Assertion (A) and Reason (R) and select the most appropriate option.

(A): A woman has normal ovulation and fertilisation, but the zygote is expelled without attaching to the uterine wall because implantation has failed.

(R): Implantation is the process by which the blastocyst embeds in the endometrium to establish pregnancy.

A) A is true, but R is false **B)** Both A and R are true, but R is not the correct explanation of A **C)** A is false, but R is true

D) Both A and R are true, and R is the correct explanation of A

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q357) In some plants, a meristematic tissue is found at the base of leaves or near the nodes, separating regions of a mature tissue. Which tissue is this?

A) Intercalary meristem **B)** Differentiated tissue **C)** Apical meristem **D)** Lateral meristem

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q358) How much energy will be available to the tertiary consumer if a food chain begins with 5000 J of energy at the producer level?

A) 0.5 J **B)** 500 J **C)** 5 J **D)** 50 J

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q359) Which of the following is a characteristic feature of skeletal muscle?

A) Involuntary control **B)** Non-striated appearance **C)** Striated appearance **D)** Single, central nucleus

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q360) Which of the following distinguishes the Rough Endoplasmic Reticulum from the Smooth Endoplasmic Reticulum?

A) RER stores lipids. **B)** RER has ribosomes. **C)** SER synthesizes proteins only. **D)** SER is rich in ribosomes.

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q361) Which of the following explains why most food chains do not exceed four trophic levels?

A) Producers are rare in ecosystems **B)** Only 10% of energy passes to the next level

C) Decomposers consume all organisms at higher levels **D)** Predators eat only herbivores

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q362) Which of the following is **NOT** attributed to Areolar connective tissues in the human body?

A) Filling the space inside organs **B)** Storing fat globules for insulation **C)** Supporting internal organs

D) Helping in the repair of tissues

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q363) Why is regular and controlled irrigation important for crop production?

A) It makes plants dependent on artificial watering. **B)** It reduces fertilizer use by supplying nutrients through water.

C) It ensures optimal water supply at critical growth stages. **D)** It increases soil salinity and prevents root growth.

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q364) Why do cells in a multicellular organism exhibit a wide range of shapes and sizes?

A) Their structure is adapted to perform specific functions.

B) Variation in cell shape and size occurs without any functional significance.

C) Cell shape is random and unrelated to function. **D)** Each cell type contains a different genome in the same organism.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q365) Which tissue in plants is primarily responsible for storage and basic metabolic functions in mature organs?

A) Sclerenchyma **B)** Parenchyma **C)** Collenchyma **D)** Epidermis

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q366) Choose a term to complete analogy.

Axon : Single long process :: Dendrite : _____

A) Many short, branched parts **B)** Protective connective tissue **C)** Cell body with nucleus **D)** Rapid stimulus transmission

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q367) Which characteristic best describes parenchyma?

A) Cells arranged in a single layer with guarding function.

B) Living cells with thin walls, capable of division in young tissues. **C)** Rigid cells that withstand bending.

D) Dead at maturity with thick secondary walls.

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q368) Which muscular tissue in animals is responsible for voluntary movements and is attached to bones?

A) Tendon **B)** Unstriated (smooth) muscle **C)** Striated (skeletal) muscle **D)** Cardiac muscle

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q369) **Which of the following tissues is responsible for limb and body movement?**

A) Epithelial tissue **B)** Cardiac muscle **C)** Smooth muscle **D)** Skeletal muscle

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q370) What distinguishes broilers from layers in poultry farming?

A) Layers are raised for meat; broilers for eggs **B)** Both are raised equally for meat and eggs **C)** Only broilers lay eggs

D) Broilers are raised for meat; layers for eggs

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q371) Meristematic tissue contributes to plant growth because its cells:

A) Have thick secondary walls **B)** Continuously divide throughout life **C)** Contain large central vacuoles

D) Are permanently differentiated

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q372) Food chains generally do not extend beyond four trophic levels because:

A) carnivores feed only on producers **B)** decomposers outnumber consumers **C)** of limited number of producers

D) of large loss of energy at each level

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q373) Which of the following epithelial tissues is specialised for absorption and lines the inner surface of the intestine?

A) Columnar epithelium **B)** Cuboidal epithelium **C)** Stratified epithelium **D)** Ciliated epithelium

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q374) Choose a term to complete the analogy.

Apical Meristem : Length :: Lateral Meristem : _____

A) Tips of stems **B)** Nodes **C)** Girth **D)** Cell differentiation only

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q375) Choose a term to pair the analogy.

Nucleus : Eukaryotic Cell :: Nucleoid : _____

A) Multicellular Organism **B)** Plant Cell **C)** Prokaryotic Cell **D)** Animal Cell

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q376) Which of the following correctly explains why veins have valves but arteries do not?

A) Veins carry blood at low pressure and need valves to prevent backflow

B) Veins carry blood under high pressure, so valves prevent it from bursting

C) Arteries carry oxygenated blood and do not need valves

D) Arteries are located deep inside the body and are supported by muscles

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q377) Which of the following is NOT a factor responsible for grain storage losses?

A) High-quality packaging **B)** Inadequate moisture and temperature **C)** Fungi and bacteria **D)** Rodents

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q378) Which of the following best describes the primary function of connective tissue?

A) Producing force through contraction and relaxation. **B)** Absorbing digested nutrients across epithelial surfaces.

C) Conducting electrical impulses for rapid communication.

D) Supporting and providing structural framework to body parts.

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q379) A gardener buried five different substances in a garden: banana peels, newspaper, cotton cloth, plastic bottle and metal spoon. How many of the substances he buried were biodegradable?

A) Four **B)** Two **C)** Three **D)** One

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q380) The principal function of the plasma membrane is to:

A) Regulate movement of substances across the cell. **B)** Store the cell's genetic material.

C) Serve as the site of ATP synthesis. **D)** Provide rigid support and prevent cell swelling.

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q381) Where in the body would you expect to find a single layer of extremely thin, flat epithelial cells designed for the transport of substances (gases) across a delicate lining?

A) Lining of kidney tubules, for mechanical support **B)** Inner lining of the intestine, for absorption

C) Lining of lung alveoli, for gas exchange **D)** Skin, to prevent wear and tear

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q382) Why is weed management important in crop production?

A) Weeds compete with crops for resources. **B)** Weeds supply nutrients to crops. **C)** Weeds increase crop yield.

D) Weeds have no impact on crops.

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q383) Which of the following is correct regarding the male and female gamete?

A) Male gamete is small and motile; and female gamete is larger and non-motile.

B) Male gamete is large and motile; and female gamete is smaller and non-motile.

C) Male gamete is large and non-motile; and female gamete is smaller and motile.

D) Male gamete is small and non-motile; and female gamete is larger and motile.

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q384) Complete the analogy.

Kharif : Rainy season :: Rabi :_____

A) Summer season **B)** Winter season **C)** Autumn season **D)** Spring season

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q385) What is formed when the male germ-cell from the pollen grain fuses with the female gamete in flowering plants?

A) Pollen grain **B)** Zygote **C)** Stigma **D)** Ovule

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q386) Blood pressure is recorded as two values: _____ pressure, measured when the heart contracts, and _____ pressure, measured when the heart relaxes.

A) Systolic; Diastolic **B)** Diastolic; Venous **C)** Maximum; Systolic **D)** Diastolic; Systolic

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q387) What is the main purpose of hybridisation in crop improvement programmes?

A) To produce genetically identical plants through cloning

B) To combine desirable traits by crossing genetically dissimilar plants

C) To grow crops without using fertilizers or irrigation **D)** To increase the natural mutation rate in crop plants

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q388) The rough endoplasmic reticulum is associated with:

A) Lipid synthesis **B)** DNA replication **C)** Protein synthesis **D)** Enzyme breakdown

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q389) What is the primary function of the axon in a neuron?

A) Store neurotransmitters for later use. **B)** Provide structural support to the neuron.

C) Transmit electrical impulses to other neurons or target organs. **D)** Supply nutrients to the cell body.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q390) Which of the following describes a key function of lysosomal enzymes?

A) Breaking down proteins, lipids, and carbohydrates **B)** Creating new proteins for the cell

C) Transporting molecules across the membrane **D)** Storing genetic material

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q391) Which of the following is a function of the Golgi apparatus?

A) Photosynthesis **B)** Protein synthesis **C)** ATP production **D)** Packaging and secretion of materials

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q392) Which connective tissue forms the framework of the external ear, tip of the nose, and the ends of long bones?

A) Areolar tissue **B)** Bone **C)** Cartilage **D)** Adipose tissue

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q393) Which part of a plant cell helps it stay intact and prevents bursting in a hypotonic environment?

A) Vacuole **B)** Cytoplasm **C)** Cell wall made of cellulose **D)** Cell membrane

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q394) Which of the following correctly differentiates ligament from tendon in human connective tissue?

A) Ligament connects bone to bone; tendon connects muscle to bone

B) Tendon is elastic, while ligament is completely non-elastic

C) Ligament connects muscle to bone, while tendon connects bone to bone

D) Both ligament and tendon connect bones to bones but differ in color

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q395) The principal role of the large central vacuole in mature plant cells is to:

A) Control cell division and heredity. **B)** Carry out cellular respiration to produce energy. **C)** Synthesize proteins for export.

D) Store cell sap and help maintain turgor pressure.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q396) What unique feature allows mitochondria to produce some of their own proteins?

A) They contain their own DNA and ribosomes. **B)** They have two membranes. **C)** They produce ATP.

D) Their outer membrane is porous.

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q397) The quality and taste of honey mainly depends on which of the following factors?

A) Age of the bees **B)** Type of beehive **C)** Type and availability of flowers **D)** Number of worker bees in the colony

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q398) What is the importance of sexual reproduction in creating variation in plant populations?

A) It speeds up DNA copying errors. **B)** It prevents any variations from occurring. **C)** It copies DNA without any changes.

D) It mixes DNA from two parents to create new variations.

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q399) Choose a term to complete the analogy.

Prokaryotic cell : Small and simple :: Eukaryotic cell : _____

A) Same size and complexity **B)** Irregular and simple **C)** Smaller and simple **D)** Larger and complex

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q400) Which statement most accurately describes why food webs provide greater stability to an ecosystem compared to single food chains?

A) They involve only producers. **B)** They have fewer organisms. **C)** They recycle only energy.

D) They allow multiple feeding relationships, ensuring balance.

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q401) Why are xylem and phloem called "complex permanent tissues"?

A) They consist of only one type of cell for performing a single function.

B) They are temporary in function and disappear after growth.

C) They contain different types of cells working together for transport. **D)** Their cells can divide and form new tissues.

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q402) Which function is mainly performed by complex permanent tissues in plants?

A) Storage of food **B)** Transport of water and food **C)** Photosynthesis **D)** Protection of plant body

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q403) Meristematic tissue in plants is mainly characterized by:

A) Permanent storage cells. **B)** Mature cells with thick walls. **C)** Small, actively dividing cells with thin walls.

D) Large, highly vacuolated cells.

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q404) Which of the following correctly represents the changes that occur in the floral parts after fertilisation?

A) The stigma and ovary both develop into fruits **B)** All floral parts persist and develop into fruits

C) The ovary develops into a fruit while the ovule becomes a seed

D) The ovule develops into a fruit and the ovary becomes a seed

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q405) The urine formed in the kidneys passes through the _____ into the ____ where it is stored until excreted.

A) urinary bladder; urethra **B)** ureters; urinary bladder **C)** nephrons; urethra **D)** urethra; ureters

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q406) Which of the following best describes the primary function of nervous tissue?

A) Absorption of nutrients **B)** Transmission of electrical impulses **C)** Providing support and binding

D) Contraction for movement

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q407) After fertilisation in humans, where does the growth and development of the embryo mainly occur?

A) Fallopian tube **B)** Vagina **C)** Uterus **D)** Ovary

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q408) A student notices that Hydra creates new organisms by developing small protrusions on its body, which eventually detach. What kind of reproduction is this?

A) Parthenogenesis **B)** Regeneration **C)** Budding **D)** Sexual reproduction

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q409) Select a term to complete the given analogy.

Binary Fission : Bacteria :: Fragmentation : _____

A) Birds **B)** Algae **C)** Mammals **D)** Insects

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q410) Which of the following factor helps in increasing crop yield?

A) Using only rainwater for irrigation **B)** Using poor seeds without irrigation **C)** Ignoring crop protection

D) Using quality seeds, irrigation, fertilisers, crop protection

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q411) Crossbreeding in cattle is done to achieve _____.

A) Longer lactation and disease resistance **B)** Excessive draught power **C)** Reduced milk production

D) Only higher milk fat content

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q412) The process wherein the body of the living organism splits into two exactly identical halves is termed as _____.

A) budding **B)** fragmentation **C)** binary fission **D)** multiple fission

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q413) Choose a term to complete the analogy.

Skeletal Muscle : Voluntary :: Cardiac Muscle : _____

A) Multinucleate **B)** Involuntary **C)** Striated **D)** Unbranched

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: B**

Q414) What is the primary function of the cell wall in plant cells?

A) Control cellular respiration **B)** Transport nutrients **C)** Control cell division **D)** Provide rigidity and protection

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q415) Which group of organisms is responsible for breaking down dead bodies and waste into simpler substances that return to the soil?

A) Decomposers **B)** Consumers **C)** Producers **D)** Parasite only

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q416) What factor primarily influences the quality and quantity of honey produced in bee farming?

A) The size of the beehive used only **B)** Flower types and their abundance for nectar gathering **C)** The colour of the bees **D)** The temperature inside the beehive alone

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q417) The female reproductive part of the flower is known as:

A) sepals **B)** stamens **C)** petals **D)** pistil

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q418) Which structure is present in plant cells but absent in animal cells?

A) Cell wall **B)** Mitochondria **C)** Endoplasmic reticulum **D)** Cytoplasm

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q419) Plastids, which are found only in plant cells, help in:

A) photosynthesis and storage **B)** excretion **C)** transport **D)** respiration

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q420) What is the main purpose of balanced nutrient management in crops?

A) To eliminate water requirements **B)** To optimize soil fertility and crop yield **C)** To promote weed growth **D)** To maximize pesticide use

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q421) Which of the following is a complex permanent tissue responsible for the transport of water in plants?

A) Collenchyma **B)** Xylem **C)** Parenchyma **D)** Phloem

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q422) Which of the following practices directly contributes to crop yield improvement?

A) Delayed harvest **B)** Planting the same crop year after year **C)** Timely irrigation and fertilization **D)** Ignoring pest signs

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q423) Sexual maturation in humans results in:

A) Loss of reproductive capacity and growth **B)** Development of reproductive organs and secondary sexual characteristics **C)** Decrease in hormone levels and loss of reproductive organs **D)** Only physical growth

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q424) Lysosomes are also known as:

A) Control centers **B)** Energy powerhouses **C)** Suicidal bags of the cell **D)** Protein factories

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Q425) Which organelle is known as the 'powerhouse of the cell' because it performs aerobic respiration?

A) Lysosome **B)** Nucleus **C)** Mitochondria **D)** Ribosome

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: C**

Environmental Science (Q-11)

Q426) Which of the following best explains why humans have the highest concentration of harmful chemicals in their bodies?

A) They eat large quantities of contaminated animal products

B) They consume more pesticides directly than other organisms **C)** They are exposed to chemicals through air and water

D) They occupy the top trophic level in food chains

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q427) When chemicals such as pesticides enter food chains and get more concentrated at higher levels, the process is known as ____.

A) Biodegradation **B)** Photosynthesis **C)** Eutrophication **D)** Biological magnification

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q428) Why is biological magnification a health risk even with limited pesticide use?

A) Non-biodegradable pesticides are immediately excreted and do not persist

B) Pesticides affect only soil microorganisms, not higher organisms

C) Non-biodegradable pesticides accumulate at successive trophic levels

D) The toxicity of pesticides decreases as they move up the food chain

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q429) In a dug well, water is collected from:

A) Neighbouring strata **B)** Deep strata **C)** Water-bearing strata **D)** Impervious strata

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q430) Which of the following combinations best explains why waste generation has become a major environmental issue in modern societies?

A) Government regulations that limit the collection of biodegradable waste and its recycling

B) Increased use of biodegradable materials and better disposal habits

C) Reduced use of packaging materials and enhanced recycling

D) Improved lifestyles leading to more disposable and non-biodegradable materials

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q431) Even though people eat different types of food, they may still absorb harmful chemical residues. This is because _____.

A) Food grains absorb only useful chemicals from soil **B)** All plants produce natural pesticides

C) Chemicals persist and accumulate via biomagnification **D)** Pesticides are completely destroyed during cooking

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: C**

Q432) Which statement correctly explains the role of decomposers in an ecosystem?

A) They recycle nutrients by breaking down dead organisms into simple substances

B) They convert inorganic substances into complex organic compounds **C)** They directly provide food to carnivores

D) They absorb sunlight to produce energy-rich compounds

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q433) The primary purpose of check-dams in irrigation is to _____.

A) To increase groundwater by storing rainwater **B)** To lower the quality of the soil **C)** To worsen erosion of top soil

D) To channel river water for factory operations

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q434) If decomposers were removed from an ecosystem, what would be the most immediate consequence?

A) Rapid increase in herbivore population **B)** Decrease in sunlight reaching producers

C) Increased plant growth due to more dead organic matter

D) Accumulation of waste and dead matter, disrupting nutrient cycling

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q435) In a school canteen, students notice that many plastic cups and plates are thrown away daily. To solve this problem, they suggest switching to steel plates and reusable water bottles. This action mainly promotes _____.

A) More plastic recycling **B)** Faster waste collection **C)** Better packaging design **D)** Reduction and reuse of waste

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: D**

Q436) Which of the following can happen when non-biodegradable waste is dumped into water bodies?

A) It pollutes the water and harms aquatic life. **B)** It helps aquatic plants grow. **C)** It cleans the water.

D) It quickly dissolves in water.

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Government Policies & Scheme (Q-9)

Q437) Under the new scheme for first-time Women, SC and ST entrepreneurs, announced in the Union Budget 2025-26, what is the **maximum loan amount** that can be availed?

A) ₹1 crore **B)** ₹2 crore **C)** ₹5 crore **D)** ₹50 lakh

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

Q438) The Rural Infrastructure Development Fund (RIDF) was established in India in which of the following years?

A) 2002–03 **B)** 2000–01 **C)** 1993–94 **D)** 1995–96

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q439) What is the primary objective of the 'Aapki Punji Aapka Adhikar' campaign, launched in 2025?

A) Expanding micro-credit access for small businesses and self-help groups across rural districts

B) Digitizing all personal banking accounts under the Jan Dhan and UPI framework for transparency

C) Facilitating speedy settlement and rightful return of unclaimed financial assets to legitimate claimants

D) Encouraging citizens to invest in long-term pension and insurance instruments for wealth creation

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q440) The NAVYA (Nurturing Aspirations through Vocational Training for Young Adolescent Girls) initiative, launched on 24 June 2025, is for which age group?

A) 12 - 14 years **B)** 14 - 16 years **C)** 16 - 18 years **D)** 18 - 20 years

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q441) Under the 2018 amendments in Foreign Direct Investment (FDI) policy approved by the Cabinet, the maximum Foreign Direct Investment (FDI) permitted under the automatic route in Single Brand Retail Trading (SBRT) is _____

A) 100% **B)** 40% **C)** 60% **D)** 85%

09/03/2026 12:45 PM - 2:15 PM **Correct Ans: A**

Q442) Which of the following initiatives was introduced in India during Union budget 2020-21 to support farmers by enabling efficient air transportation of agricultural produce across domestic and international routes?

A) Rural Export Enhancement Initiative **B)** National Farm Logistics Mission **C)** Krishi Air Connect Programme

D) Krishi Udan Scheme

06/03/2026 9:00 AM - 10:30 AM **Correct Ans: D**

Q443) What is the focus of the 6G research project at IIIT-Naya Raipur funded under DST's TTDF scheme?

A) Expanding smartphone production facilities **B)** Creating an AI-driven telecom cloud platform

C) Building a 50-qubit quantum computer **D)** Developing a prototype for 6G cell-free communication

10/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q444) The National Sports Governance Bill, 2025, was introduced to achieve which of the following objective?

A) To establish rules for recognizing and regulating sports bodies **B)** To improve physical and sports education in schools

C) To launch a national-level talent hunt program **D)** To increase budget allocation for Olympic sports

09/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**

Q445) Which of the following is the primary objective of Mission Sudarshan Chakra, launched by India in August 2025?

A) Support agricultural innovation **B)** Create a powerful defence weapon system **C)** Promote economic reform

D) Strengthen civil aviation

10/03/2026 12:45 PM - 2:15 PM **Correct Ans: B**

SelectionWay

Static GK (Q-5)

Q446) Which of the following dimensions is NOT directly included in the Human Development Index (HDI)?

A) A long and healthy life **B)** A decent standard of living **C)** Human security **D)** Being knowledgeable

06/03/2026 12:45 PM - 2:15 PM **Correct Ans: C**

Q447) On which date was Engineers' Day observed in India in 2025?

A) 14 September **B)** 15 September **C)** 16 September **D)** 13 September

09/03/2026 9:00 AM - 10:30 AM **Correct Ans: B**

Q448) Which distinguishing feature primarily differentiates a biosphere reserve from a national park in India, as per conservation and management objectives?

A) Biosphere Reserves (BRs) allow research, tourism, and local livelihoods in designated buffer and transition zones.

B) Biosphere Reserves (BRs) covers a smaller geographical area than national parks.

C) Biosphere Reserves (BRs) focuses exclusively on protection of wild animals.

D) Biosphere Reserves (BRs) does not aim at conservation of biological diversity.

10/03/2026 9:00 AM - 10:30 AM **Correct Ans: A**

Q449) India celebrated Engineers Day on 15 Sep 2025 with the theme "Deep Tech & Engineering Excellence" in memory of whom?

A) A.P.J. Abdul Kalam **B)** Jagadish Chandra Bose **C)** Srinivasa Ramanujan **D)** Sir M. Visvesvaraya

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: D**

Q450) In which of the following years was the 'Indus Water Treaty' signed between India and Pakistan?

A) 1960 **B)** 1947 **C)** 1955 **D)** 1963

06/03/2026 4:30 PM - 6:00 PM **Correct Ans: A**